An LLM-Based Motivation-Aware Framework For AI Coaching For Behaviour Change

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Abstract

Digital health coaches can deliver affordable, accessible behaviour coaching to broad populations. While prior research in motivational interviewing (MI) systems has emphasised emotional support strategies, it has overlooked the role of goal-setting behaviours in achieving successful coaching outcomes. We propose a coaching framework that provides motivational support for ambivalent clients while employing planning-focused strategies for action-ready clients. The user study evaluated our framework against a baseline system for promoting physical activity. Results showed significant increases in users' readiness to change after interacting with our motivational coaching chatbots. Participants perceived our agent as more adherent to MI principles and more effective at planning than the baseline. Qualitative analysis revealed that users who agreed with the agent on a change plan experienced the greatest increases in intrinsic motivation. Finally, we examine how these findings align with expert opinions that informed our system design principles and discuss implications for LLM-based health coaching.

CCS Concepts

• **Human-centered computing → Empirical studies in HCI**; **Interactive systems and tools**; **User studies**; • **Computing methodologies → Natural language processing**.

Keywords

LLMs; Conversational Agents; Coaching Agents; Natural Language Processing; Human-AI Interaction; Behaviour Change; Physical Activity; Motivational Interviewing; Motivation

ACM Reference Format:
Van Hoang, Eoin Rogers, and Robert Ross. 2026. An LLM-Based Motivation-Aware Framework For AI Coaching For Behaviour Change. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), April 13–17, 2026, Barcelona, Spain.* ACM, New York, NY, USA, 24 pages. https://doi.org/10.1145/3772318.3791123

1 Introduction

Health coaching represents a collaborative, goal-driven partnership that empowers clients to take charge of their health through guided discovery and personal development [52]. Research demonstrates that adopting positive health behaviours yields substantial benefits for both physical and mental wellbeing, with behavioural interventions consistently showing greater cost-effectiveness and lower risk compared to medication [44]. Helping clients change is an important function of health providers. However, behaviour change is challenging and while in-person coaching by trained professionals remains the gold standard, this approach is both costly and difficult to scale. Therefore, digital health coaches have emerged as a promising solution for delivering behavioural interventions. These conversational agents, or chatbots, use natural language interactions to provide affordable, and accessible health behaviour coaching services to a wide audience [66].

Motivational interviewing (MI) is a client-centred evidence-based psychotherapeutic approach designed to strengthen intrinsic motivation for behaviour change [47]. Within health coaching contexts, MI functions as a cognitive intervention targeting human intrinsic values and motivations, as opposed to techniques grounded in behavioural theories that focus on planning and realistic goal-setting [28, 33]. MI coaching agents have demonstrated effectiveness across multiple domains [44, 55, 65], including eating habits [60], alcohol use [50], smoking cessation [11], and physical activity [36]. Early digital health coaches predominantly relied on rule-based or retrieval-based systems that lacked conversational flexibility, significantly limiting their effectiveness. Large Language Models (LLMs) have emerged as a transformative technology with substantial potential for psychotherapy applications [70]. Despite being in their technological infancy, consumers have already begun using non-clinical LLMs as informal counselling tools [10, 18]. Recent advances in Natural Language Processing have successfully applied LLMs to MI systems, dramatically enhancing conversational flexibility and enabling agents to deliver personalized responses tailored to individual user needs [35, 45, 71].

Prior research on MI agents [11, 45, 71] has focused heavily on strategies that support users' emotions, neglecting the role of goal-setting behaviours in strengthening the therapeutic relationship. This limitation is significant as MI can be counterproductive for highly motivated clients, indicating the need for motivation-tailored interventions [20]. While recent work has attempted to integrate both strategy types [12, 35], these agents tend to offer unsolicited problem-solving advice.

Counselling requires balancing two competing goals: building relationships with clients and promoting their behavioural change. While excessive challenging may rupture therapeutic bonds, over-emphasising relationships could prevent change. The therapeutic

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CHI '26, Barcelona, Spain
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ACM ISBN 979-8-4007-2278-3/2026/04
https://doi.org/10.1145/3772318.3791123

zone of proximal development (TZPD) is a framework where supporting strategies provide safety while challenging actions encourage manageable risk-taking [59]. Inspired by this, we implement a system that personalises strategies based on users' readiness for change. While prior works primarily focus on emotional states [37, 38, 43], our approach relies on user motivational states for prediction. This distinction is crucial as clients may exhibit happy yet unmotivated states (e.g., *"I feel good with how things are going. I see no reason to change my behaviours"*) or angry yet motivated states (e.g., *"I hate myself for not stopping drinking before, but I want to try again."*)

Our framework leverages MI coding system. In MI, client utterances are classified as expressing or opposing change, along with commitment strength. We use this to gauge clients' motivation towards their goals, informing whether subsequent actions should prioritise building rapport or pushing for change. The coaching agent can provide support for low-motivated users and resorts to planning strategies otherwise. We demonstrate our framework using physical activity promotion as a low-risk test case, though it can be adapted to other behavioural change domains by modifying prompts and adding domain-specific knowledge. We conducted a mixed-methods evaluation using a single-session study design, recruiting 140 participants via the Prolific online research platform. The effectiveness of our system is compared against a baseline system which offers limited motivational and emotional support while primarily focusing on task planning.

The key contributions of this work are as follows:

(1) We propose an adaptive coaching framework[1] that adapts strategies to users' motivational states, grounded in established psychological theories and informed by human coaching literature. Compared to the baseline system, our agent exhibits greater adherence to MI best practices: using less persuasive language without permission, withholding unsolicited advice, and utilising a wider range of MI behaviours.
(2) Empirical findings from a user study demonstrate that even brief interactions with AI motivational coaches significantly increase users' intrinsic motivation for behavioral change.
(3) We explore the behaviours that contribute to successful coaching experiences, providing qualitative insights into user preferences and expectations that may conflict with established MI principles. These findings can inform the future design of behaviour change support system tailored to different user profiles.

2 Related Works

2.1 Insights from Health Science and Practice

Face-to-face coaching delivered by an expert human coach remains the "gold standard" that automated systems should strive for [5]. Research in human coaches indicates that health coaching interventional components of goal-setting and motivational interviewing (MI) are common themes [53]. Goal-setting enables clients to self-monitor behaviours and create personalised health plans, while MI strategies help them realise their change goals.

Rutjes et al. [61] conducted three qualitative studies exploring health coaches' perspectives on the role and requirements of technology in their practice. Their first study examined coaching processes and factors that influence successful coaching sessions. Coaches revealed that success goes beyond measurable goals and activities. Increased awareness on the impact of a certain diet or behaviour on health is considered as a valuable benefit. Additional key success factors include establishing strong coach-client relationships, personalising advice to individual circumstances, fostering clients' intrinsic motivation, and setting specific, achievable short-term goals that enhance motivation and self-efficacy. In their studies, Rutjes et al. [61] explored the coaches' perceptions and attitudes towards technology integration within coaching processes. Technology can assist with detecting emotions, non-verbal cues, and, most importantly, change talk, which is the verbal signals indicating clients' willingness to change a certain behaviour [47].

Similarly, Jörke et al. [35] conducted formative semi-structured interviews with 22 health experts to understand coaching strategies and potential applications of LLMs in health coaching contexts. Experts came from diverse professions including health coach, fitness instructor, physical therapist, and behavioural scientist. Experts recommended a facilitative rather than prescriptive approach, avoiding unsolicited advice or problem-solving. Advising through assumptions can damage credibility and trust. Furthermore, facilitation empowers clients and fosters self-efficacy, enabling them to develop their own problem-solving capabilities. Notably, several experts independently cite MI as a guiding framework for such an approach. Nevertheless, offering information during coaching remains appropriate when sufficient client information has been gathered to ensure that advice aligns with their needs, abilities, and resources. Experts emphasise that building rapport and trust is essential, as health coaching is fundamentally a relational partnership where coach and client collaborate toward shared goals.

2.2 Psychological Theories

2.2.1 Motivational Interviewing. Psychology has long studied how and why people resist the pressures to conform. In counselling, resistance proves to be a serious issue, limiting its effectiveness [76]. MI is an evidence-based therapeutic approach that aims to reduce resistance and enhance clients' intrinsic motivation for behavioural change [47]. Core MI strategies include employing reflective listening, asking open-ended questions, offering genuine affirmations of clients' strengths, and emphasizing their autonomy in the change process. MI unfolds through four sequential phases, each serving a distinct purpose: engaging (building a collaborative therapeutic relationship), focusing (identifying specific areas for change), evoking (strengthening intrinsic motivation), and planning (developing specific plans). The main goal of MI is to evoke "change talk" – clients' verbal expressions of commitment to change, which has been identified as a key predictor of actual behavioural change and therefore represents a crucial focus for counsellors [4].

2.2.2 Motivational Interviewing Treatment Integrity (MITI) guideline. MITI [48] is a standardised behavioral coding system designed to assess the quality and fidelity of Motivational Interviewing (MI) practice. It tracks and codes specific counsellor behaviours that are central to MI practice. There are 9 behaviours in total: Give

[1]Data and code can be found at: https://github.com/VanHoang85/mi-health-coach

Information, Persuade with Permission, Questions, Reflection, Emphasize Autonomy, Affirm, Seek Collaboration, Persuade without Permission, and Confront. Widely regarded as the gold standard for MI fidelity assessment, the MITI is used in clinical training, supervision, research studies, and quality assurance programs to help practitioners develop and maintain their MI skills.

2.2.3 The Transtheoretical Model (TTM). Also known as the Stages of Change Model [58], it describes how people move through different stages when changing health-related behaviours such as quitting smoking, exercising, or changing diet. The model identifies five main stages: (1) Precontemplation: Not yet considering change, unaware or in denial about the problem; (2) Contemplation: Aware of the problem and thinking about change, but not committed yet; (3) Preparation: Ready to take action soon; making small steps or plans; (4) Action: Actively modifying behaviour and environment (first 6 months); and (5) Maintenance: Sustaining the new behaviour and preventing relapse (after 6 months).

A key insight of the TTM is that change is not linear: people often move back and forth between stages and encountering obstacles before sustaining new habits. The model also emphasises that different interventions work better at different stages, so knowing a person's current stage allows for more targeted and effective assistance. It's widely used in health promotion, addiction treatment, psychotherapy, and public health programs.

2.2.4 The EPITOME Model. Empathy is fundamental to therapeutic relationships and enhances communication effectiveness across social interactions [49]. This crucial skill correlates with better outcomes in human-centred fields such as medical care and psychotherapy [19], making measures of empathy valuable indicators of therapeutic quality. In MI, where counsellors facilitate clients' intrinsic motivation for change rather than prescribe solutions, empathy serves as both a foundational element and a key indicator of therapist effectiveness [78].

`EPITOME`, a computational framework Proposed by Sharma et al. [63], measures empathy in peer-to-peer support conversations exclusively from text. The model computes empathy scores using messages from both support seekers and providers, not providers alone. `EPITOME` evaluates empathy across three distinct dimensions:

- Emotional Reactions: expressing emotions such as warmth, compassion, and concern.
- Interpretations: communicating an understanding of feelings and experiences inferred from users' texts.
- Explorations: probing into feelings and experiences beyond those explicitly mentioned by the user.

Each dimension is scored on a three-point scale: (0) no communication, (1) weak communication, or (2) strong communication. These scores reflect the extent to which participants demonstrate communicative effort in their messages. Total scores range from 0 (absence of empathic communication) to 6 (strong empathic communication across all dimensions).

2.3 Conversational Systems as Health Coaches

Conversational systems, either text-based chatbots or embodied avatars, aim to replicate face-to-face interaction with human experts, the gold standard for coaching [6], by leveraging NLP and AI technology. As such, many conversational AI coaching agents utilise principles and strategies from human coaching, emulating the behaviours of health experts. In a meta analysis of conversational systems on lifestyle behaviour changes, Singh et al. [66] found that the most commonly integrated behaviour change interventions include goal-setting, self-monitoring, reviews of goals, problem-solving barriers, motivation, feedback and peer support. Additionally, the effect size for behaviour change outcomes is found to be small to medium.

Due to technological constraints, most coaching agents rely on scripted, rule-based frameworks, preventing users from engaging in natural, free-flowing conversations. To assist with goal-setting, some conversational agents can also use external resources such as tracked data from wearable devices to learn about the users' behaviours and then suggest suitable goals and provide educational information [42, 74].

Created by Maher et al. [42], Paola, the digital coach, guides users through 11 weekly check-ins regarding their daily steps and dietary patterns, and assists them in setting goals for upcoming weeks using their activity logs. Additionally, Paola has access to an external database developed with health scientists so it can answer users' questions about food and exercises. Similarly, To et al. [74] developed Ida, a physical activity coach which uses activity data tracked by Fitbit to personalise daily goals. To enhance user motivation, Ida delivers encouraging messages and achievement compliments via Facebook Messenger. The digital coach by Gupta et al. [28], instead of suggesting daily goals for users based on guidelines, assumes that users come to the coaching system with specific goals in mind and mainly need assistance in formulating these goals' SMART attributes[2]. Their agent analyses each user message to identify existing goal attributes, then asks questions to elicit any missing SMART components.

2.4 MI Agents as Health Coaches

Contrary to the systems that focus on setting goals, MI agents target building intrinsic motivation and reducing resistance to change. These agents have demonstrated effectiveness across multiple domains [44, 55, 65], including eating habits [60], alcohol use [50], smoking cessation [11], and physical activity [36]. Traditionally, the MI conversational flows are all scripted or semi-scripted, restricting users' natural language expressions and thus limiting the effectiveness [11, 21, 51, 54].

While early AI health coaches relied on rule- or retrieval-based systems, recent developments leverage LLMs for MI coaching. Meyer and Elsweiler [45] simulated GPT-4 to act as an MI coaching agent to interact with participants on Prolific[3]. They constructed only one prompt format for the entire conversation and the coaching strategies were predicted using a classifier on the current user utterance. Many participants expressed frustration that the MI-adapted version did not provide advice or factual information, even though they had been informed in advance that this would be the case. Xie et al. [79] proposed a framework that learns MI strategies through inductive reasoning, using a retriever to match dialogue contexts (including clients' states and therapists' strategies) for response

[2]SMART = Specific, Measurable, Achievable, Realistic, and Timely.
[3]https://prolific.com/

generation. The agent by Steenstra et al. [71] utilised prompt engineering that integrated MI principles and was evaluated by engaging 8 MI-expert participants role-played as clients. Initial findings suggested that the LLM-powered virtual agent matches human counsellors' empathetic and adaptive conversational skills. However, it was universally found to over-rely on complex reflections and missed opportunities both to ask questions to facilitate deeper engagement, and to set up a plan once a client agrees to make some incremental change: the latter aligns with Meyer and Elsweiler [45]'s findings on limited practical guidance. An expert found that the agent struggled to adapt its approach based on clients' different stages of change. In summary, while autonomous LLMs enhance conversational flexibility, they compromise both controllability and explainability since their underlying coaching strategies remain unclear in the generation process.

Adopting a phase-based approach, the coaching system by Jörke et al. [35] follows a fixed linear order of phases regardless of user readiness. Their design uses a single prompt per phase that specifies behaviours of the agent. Each phase serves a specific purpose, for instance, to ask about past experiences, or to gather information on obstacles for change. A classifier is called after each user utterance to predict whether the agent should move to the next phase, while another classifier predicts MI strategies for the agent's next turn. Despite their efforts to restrict MI-inconsistent behaviours, the agent still offers unsolicited advice too early during the coaching session. It is hypothesised that these limitations stem from LLMs' training paradigm, which primarily optimises for single-turn question answering [12, 35]. As indicated earlier, this training approach results in systems that tend to passively follow users' instructions rather than proactively engage in meaningful dialogue through strategic questioning [15].

2.5 Design Guidances for Our AI Coach

Given the relevant literature, we formulate the following design principles for our AI health coaching agent:

(1) **Dual purposes.** The coaching sessions should serve two purposes, namely fostering motivation and setting specific and achievable goals [53]. Intrinsic motivation is important and the AI coach should enhance it first before assisting users in goal-setting.
(2) **MI as the guiding framework.** MI has been cited by multiple sources as an effective approach to build intrinsic motivation and empower clients to direct their own journey of change in behaviour, rather than imposing directions on them [35, 53].
(3) **Personalised information and advice.** Unsolicited advice and premature information are highly undesirable [35, 61]. The AI coach must first thoroughly explore users' unique circumstances before offering personalised guidance.
(4) **Sufficient therapeutic connection** Coaching fundamentally depends on a relational and collaborative process. The AI coach must establish genuine rapport with users to create the trust necessary for effective behaviour change work [35, 61].
(5) **Hybrid systems.** MI coaching frameworks should integrate both the explainability of traditional systems and the natural conversational fluency of LLMs. Traditional systems, while lacking natural language expression, offer transparency in their application of counselling strategies. In contrast, LLMs can adapt well to user utterances, but autonomous, general-purpose LLMs struggle to effectively function as coaches due to limitations in their training paradigm [12], [45].
(6) **Proactive agents.** The AI coach must be capable of steering conversations strategically rather than merely responding to user requests [15]. This requires both a proactive dialogue policy and sophisticated reasoning capabilities to determine optimal strategies for achieving coaching objectives.

These design principles guide the implementation of our coaching agent. We now describe key architectural decisions that operationalise these principles.

We structure coaching sessions into distinct phase, each with a specific goal, and restrict the agent's behaviours to only those which align with each phase's purpose. This structured approach enables the agent to be more proactive while mitigating a key limitation of autonomous AI coaching agents: their training paradigm encourages passive response to user requests, often resulting in unsolicited advice rather than purposeful guidance [12, 35]. Thus, similar to traditional systems, our AI coach employs a dialogue policy which determines the next phase and coaching strategies of the conversations. However, traditional rule-based systems rely on predetermined question sequences and fixed content structures, which severely limit user expression and natural interaction. We instead provide high-level behavioral guidelines for the coaching agent. For example, exploring users' beliefs through open-ended questions such as "Why do you want to exercise more?". This hybrid approach preserves the agent's linguistic flexibility while also maintaining the ability to handle off-topic questions.

MI comprises five sequential phases: engaging, focusing, evoking, planning, and concluding. While this linear progression shows significant benefits for those expressing ambivalence, research indicates it may be less effective—and potentially counterproductive—for clients already exhibiting strong readiness for change [20]. These findings suggest the importance of tailoring coaching strategies to align with clients' presenting motivation levels. For this reason, modelling user motivational states should enable the AI coaching agent to determine appropriate global policies and local strategies. Global policies outline general goals while local strategies specify actions to achieve them. While prior works in AI coaching systems primarily focus on emotional states [37, 38, 43], our approach relies on motivational states. This distinction is crucial as clients may exhibit happy yet unmotivated states (e.g., *"I feel good with how things are going. I see no reason to change my behaviours"*) or angry yet motivated states (e.g., *"I hate myself for not stopping drinking before, but I want to try again."*)

3 MI-Aware Coaching Framework

Based on the design principles from the previous section, we propose a computational framework that can select coaching strategies depending on users' motivational states.

Figure 1 illustrates our MI-aware coaching framework, which consists of four main components: (1) the client motivational language detector, (2) the phase manager, (3) the coaching strategy

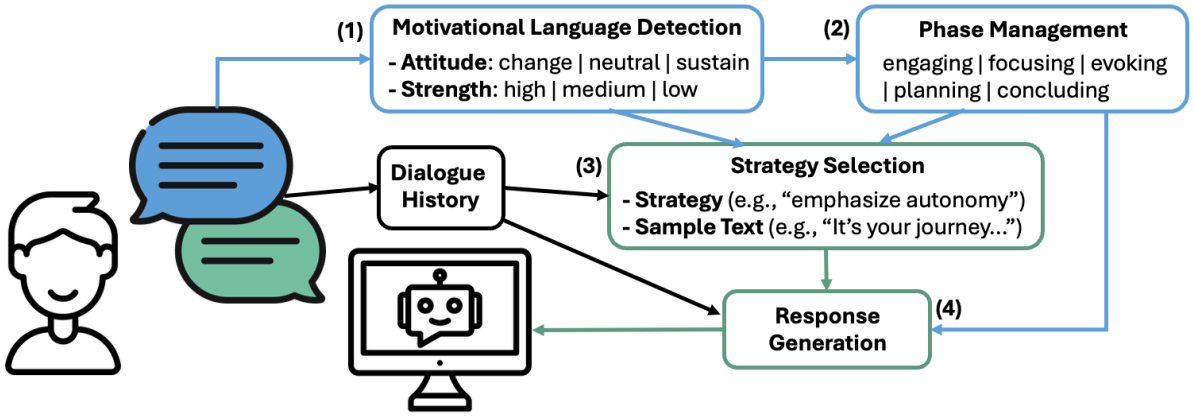


Figure 1: Overview of our MI-Aware Coaching System.

selector, and (4) the response generator. Given a dialogue history between a client and an agent with the last utterance spoken by the client, the motivational language detector (1) estimates user' levels of motivation following MI coding convention [3]. This information is then passed to the phase manager (2) to determine the next coaching phase after a certain number of turns. The MI-aware coaching agent operates within specific behaviours for each phase with the strategy selector (3) predicting the most suitable strategies and corresponding sample texts aligned with MI principles. Finally, the response generator (4) formulates reply to the client's utterance, using relevant data from other components.

To maintain focused scope and reduce system complexity, we design our coaching agent implementation for a single-session user study rather than a longitudinal study. A longitudinal approach would require additional modules to store and retrieve conversation histories across multiple sessions. Physical activity promotion is used as a low-risk case study for the framework, though it can be adapted to other behavioural change domains by modifying prompts and adding domain-specific knowledge. In the next sections, we will describe each component in detail.

3.1 Motivation Estimation from MI Dialogues

The task of predicting client motivational language can be divided into two subtasks. The first one, called the `attitude` task, is to detect the direction of motivation: whether the client is willing to change or not. The other one, called the `strength` task, is to detect the commitment level: if the client is willing to change or still shows resistance, how strong do they hold such beliefs? For example, *"What do you mean? There is no problem with my health"* is labelled as "sustain" for `attitude` and "high" for `strength` while *"I think I might try to walk more."* is annotated with "change" and "low" respectively.

For the `attitude` subtask, models receive both coach and client utterances as input, while the `strength` subtask uses only client utterances. In both cases, models predict labels for client turns. We experiment with different model families using in-context learning (ICL) and prompt engineering techniques. ICL enables LLMs to perform new tasks by including several examples as context within the prompt itself, allowing models to learn solely by "observing" these demonstrations. We refer to ICL as n-shot when there are n demonstrations in the prompt. For our experiments, we utilise dialogues from the `AnnoMI` dataset [77] as in-context examples. It comprises 133 MI conversations across 10 different topics in English, transcribed from YouTube demonstration videos and annotated by experts from the MI network[4]. The dataset is available under Public Domain License, with additional details provided in Appendix A.

We measure both task accuracy and inference time. Considering latency constraints, we choose `gpt-4o-mini` for our user study. The attitude task employs 3-shot ICL, while the strength task uses a 0-shot approach. The motivational language detector generates two key outputs for each client utterance: an `attitude` label and a `strength` label. Before passing these outputs to subsequent system components, we convert them to motivational levels on a -3 to +3 scale, where 0 represents a "neutral" attitude [22]. "High", "medium", and "low" strength labels are assigned values of 3, 2, and 1 respectively, while "change" and "sustain" attitudes determine positive or negative direction. For instance, "change-high" equals +3, while "sustain-low" equals -1. Outputs from this step are passed to (i) the phase manager to determine the next coaching phase, and (ii) the strategy selector to aid in re-ranking its candidates.

3.2 Phase-Based Dialogue Policy Management

To enhance controllability of the system, we divide the coaching sessions into five phases, namely "engaging", "focusing", "evoking", "planning", and "concluding". The phase manager adjusts the global policy by selecting phases that align with clients' current readiness levels. Phase descriptions are presented in Table 1. The final "concluding" phase is not part of the MI intervention itself but serves to facilitate a smooth conversational ending with users. A turn consists of one client utterance followed by one coach utterance.

Following engagement, the agent assesses users' perspectives during the "focusing" phase to gauge their base motivational readiness. Client readiness is assessed through motivational levels. At the end of the phase, the system uses majority voting to categorise

[4]https://motivationalinterviewing.org/

Table 1: The coaching agent's global policies and local strategies in each phase. Global policies outline general goals while local strategies specify actions to achieve them. Depending on the clients' stages of change, either "evoking" or "planning" phase might not occur.

phase	the agent's global policy	possible local strategies	number of turns
engaging	get to know clients	introduce, ask focusing questions	2 turns
focusing	explore clients' beliefs	ask focusing questions, do simple reflections, do complex reflections, affirm, emphasise autonomy	5 turns
evoking	build up motivation for change	ask focusing questions, ask evoking questions, do simple reflections, do complex reflections, affirm, emphasise autonomy, seek collaboration, give information	0-12 turns
planning	assist in formulating action plans	ask focusing questions, ask evoking questions, affirm, emphasise autonomy, seek collaboration, give information, persuade with permission	0-12 turns
concluding	wrap up the session	summarise conversations, terminate conversations	1-3 turns

client states: resistance (levels -3, -2), ambivalence (levels -1, +1), or motivation (levels +2, +3). When the predominant level is 0, the system treats clients as ambivalent. In real MI sessions, the mean duration before the therapists begin actively influencing the clients' motivation was approximately 5 minutes [20]. Our "focusing" phase mirrors this with 5 turns, where each turn (one client-coach exchange) corresponds to one minute of real-time interaction. The duration of the "evoking" and "planning" phases was established through pilot testing conducted with researchers at our institution during system development. We determined that a maximum of 12 turns per phase is sufficient to achieve each phase's core objectives.

With user readiness estimation, the agent strategically determines when to foster motivation through the "evoking" phase for resistant or ambivalent clients, or proceed directly to goal-setting in the "planning" phase for those already motivated for change. Conversely, when resistance persists throughout the session, the "planning" phase is omitted entirely. Using the user motivational modelling, the coaching agent can dynamically switch between the "evoking" and "planning" phases depending on detected user readiness levels. The "concluding" phase is initiated under two conditions: either when a concrete change plan has been formulated, or the interaction reaches 20 turns in total.

3.3 Coaching Strategy Selection Via Motivation Estimation

Instead of allowing the coaching agent unrestricted autonomy, its behaviours are constrained to those specifically aligned with each phase's defined global policy. Coaching strategies are categorised as either supporting or challenging, where supporting strategies create a safe therapeutic environment for clients, while challenging strategies encourage manageable risk-taking that pushes clients toward their change goals [59]. To maintain therapeutic rapport during the coaching process, coaches must skilfully balance these two types of strategies.

Following the Motivational Interviewing Treatment Integrity (MITI) guideline [48], the AI coaching agent employs nine MI-adherent strategies, classified into supporting and challenging. These strategies are detailed in Table 2, together with their descriptions and examples. We acknowledge that this separation between two categories is not clear-cut. For example, "Complex Reflection" might evoke in clients the desire for change. Nevertheless, our classification is based on their intended usage.

Within our framework, the "focusing" phase emphasises supporting strategies, while "evoking" and "planning" phases integrate both types to balance relationship development with coaching progress. The "engaging" and "concluding" phases follow a more structured approach with predefined behaviours. The strategies allowed for each phase are listed in Table 1.

For each turn i, the strategy selector determines local strategies s_i based on the phase's global policy p_i, client's motivational level m_i, and dialogue history h_i. A retriever selects appropriate strategies for the next conversational turn. This approach leverages the intuition that similar contexts typically benefit from similar strategies. While an alternative would be training a classifier for strategy prediction, the highly imbalanced nature of the dataset implies certain strategies would rarely be predicted. Additionally, a single client utterance can elicit diverse valid responses: Experts verified that even when predicted strategies differ from ground-truth annotations, they can still effectively align with MI principles and contexts [72]. In other words, our approach does not attempt to identify a single "best" strategy for each turn. Instead, by allowing the agent a predefined set of behaviours for each coaching phase, we work under the assumption that any of these behaviours can successfully maintain human-agent rapport throughout the coaching session. The retriever's main function is therefore to identify the most contextually appropriate candidate text. As shown in Figure 1, an example of the retrieved strategy is "emphasise autonomy" and the retrieved text can be *"It's your own journey. I can't make that decision for you."*

At turn i, candidate strategies are identified through a multi-step process. First, a dense retriever selects dialogue contexts from the database that closely match the current dialogue history. These candidates are then filtered to retain only those strategies that align with the phase's global policy. A subsequent re-ranking step ranks

Table 2: MI-adherent strategies used by our MI-aware coaching system. They are adopted from the MITI 4.2 manual [48].

Strategy	Type	Description	Example
Open/Closed Question (Focusing)	Supporting	Gather information, understand, or elicit the client's story about what they previously said.	"What are your feelings about those types of activities?"
Open/Closed Question (Evoking)	Challenging	Evoke the client's desire or need to change.	"How could making this change make your life better?"
Simple Reflection	Supporting	Capture and return to the client something that they have said.	"You're motivated to make a change that improves your energy."
Complex Reflection	Supporting	Add significant meaning or emphasis to what the client has said. They convey a deeper or richer picture of the client's statement.	"It sounds like you're somewhat interested, but I also get the feeling that you have some hesitations."
Affirm	Supporting	Say something positive or complimentary to the client.	"You came up with a lot of great ideas. Great job brainstorming today."
Seek Collaboration	Supporting	Ask for permission to offer an advice, a piece of information, or your thoughts on what they just say.	"I have some suggestions you might consider. Would you be interested in hearing them?"
Emphasise Autonomy	Supporting	Highlight clients' sense of control, freedom of choice, personal autonomy, ability or obligation to decide about their attitudes and actions.	"You're the expert on your body, so it's up to you what you choose to do."
Give Information	Challenging	Provide information to the client, explain ideas or concepts to the intervention, educate about a topic, or express a professional opinion on the client's habits or actions without persuading, advising, or warning.	"According to the UK Chief Medical Officers' Physical Activity Guidelines, adults should try to be active every day and aim for at least 150 minutes of physical activity over the week."
Persuade With Permission	Challenging	Give advice, make a suggestion, or offer choices of solutions or possible actions. Negotiate with the client on creating an action plan to achieve their goal.	"For instance, instead of sitting to read or watch TV, you could consider standing or moving around more frequently."

candidates based on the client's motivational level. For example, if a client exhibits high resistance (level -3), the retriever identifies dialogues with various resistance levels (-3, -2, -1). Re-ranking then ensures dialogues matching the client's specific resistance level (-3) appear at the top of the candidate list, providing the most contextually relevant responses.

For the retrieval step, we build a dialogue database D_{base} from the `AnnoMI` dataset (See Appendix A) and a mapping from the dialogue history to the next coaching utterance $\{D_i \rightarrow C_i\}$. The dialogue histories are stored using ChromaDB.[5] and we employ Cohere[6] as the embedding model.

3.4 Response Generation

The response generator creates replies to client utterances. The global policy corresponding to the current phase p_i serves as the system prompt. To ensure responses align with local coaching strategies, we utilise retrieval-augmented generation (RAG) approach that retrieves and integrates relevant example texts as contextual cues. For instance, when the strategy calls for "asking a focusing question," the system might retrieve a contextually appropriate cue such as *"What changed for you? Why does it feel like a chore now?"*. We further instruct the model to adapt the phrasing of retrieved cues to maintain coherence with the user's utterance.

For each turn i, we construct strategy-specific prompts using the retrieved strategies s_i and their associated sample responses r_i. The final prompt pr_i concatenates three elements: the global policy, the local strategy instructions, and the five most recent dialogue turns. The complete prompt structure is detailed in Appendix C.

4 Methodology

4.1 Study Setup

For the user study, we implement two LLM-based coaching agents: a base-MI agent and our proposed MI-aware agent (described in Section 3). The base-MI agent functions as the baseline system. It is given access to the complete dialogue history, which simply instructs GPT-4o[7] to act as a physical activity coach using MI techniques without additional prompt engineering.

We developed web interfaces for both systems using Gradio [1] and deployed them on Hugging Face Spaces[8]. Figure 2 shows

[5]https://www.trychroma.com/
[6]https://cohere.com/blog/introducing-embed-v3
[7]Version gpt-4o-2024-08-06
[8]https://huggingface.co/spaces

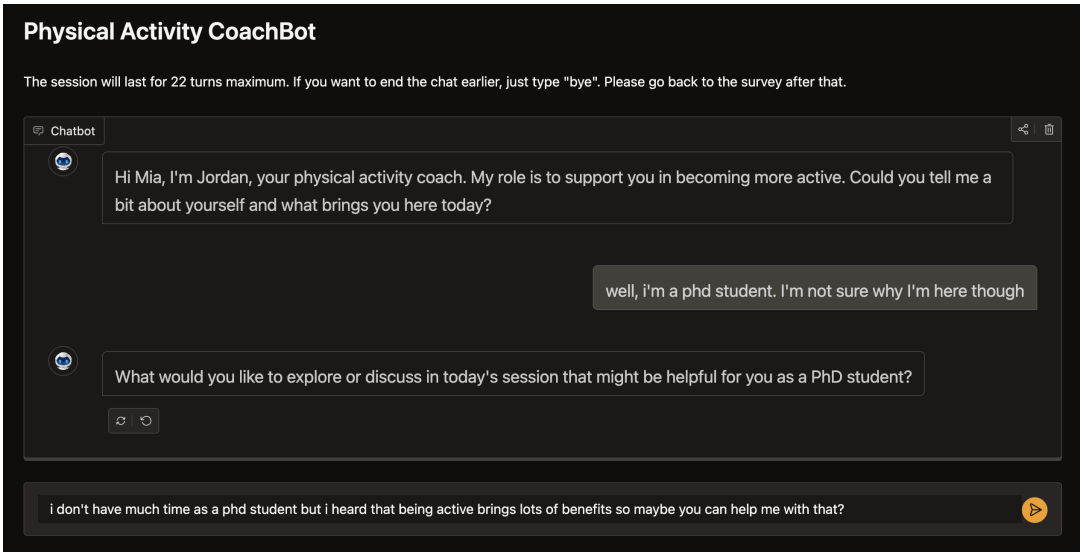


Figure 2: The chat interface between participants and the AI health coach.

the turn-based interface through which participants interact with the AI coaches. We impose a maximum limit of 22 conversational turns across all conditions, while allowing participants to terminate sessions early if desired.

We collected two types of data: questionnaire responses and human-agent interactions. Upon enrolling, participants were asked to provide informed consent before completing three questionnaires. After that, participants were randomly allocated to interact with one of two coach settings: (1) the base-MI agent, and (2) the MI-aware agent. The randomisation was double-blinded and carried out automatically using Prolific features. After the interaction, participants completed post-experimental surveys to evaluate changes in the measured variables.

4.2 Measures

The pre-experimental surveys measure participants' intention to do exercises, baseline physical activity level, and their perceived relationship with the coaching agent. For behavioural outcome measures, we collected both motivation- and user experience-related variables. The list of three dependent variables is as follows:

(1) Δ Readiness to Change – The Contemplation Ladder (CL) [7] is an instrument with descriptive labels representing different stages of readiness to change. It measures the motivation for change on a 11-point Likert scale, ranging from 0 (no thoughts of changing) to 10 (already taking actions to change). Participants indicate where they are on the ladder before and after the interaction with the coaching agent.
(2) Perception of MI – We measure user perception of MI using the Client Evaluation of Motivational Interviewing Scale (CEMI) [39–41]. The CEMI is an 11-item self-report instrument that employs a 5-point Likert scale to capture user experiences of MI interactions. The scale evaluates both relational components and technical aspects of MI. Participants rate each item from 1 (not at all) to 5 (always) following their interaction with the coaching agent. Higher total scores indicate greater consistency with established MI principles and techniques.
(3) Therapeutic Alliance – We measure the therapeutic relationship using the Working Alliance Inventory Short Form (WAI) Client Version [30]. The WAI is a 12-item self-report instrument that employs a 5-point Likert scale, with responses ranging from 1 (seldom) to 5 (always). The scale comprises three derived subscales: Goal (agreement on goals), Task (agreement on tasks and activities), and Bond (the therapeutic connection between client and coach). Participants complete it after interacting with the coaching agent. Higher scores indicate a stronger working alliance, which consistently predicts better outcomes across interventions.

We collected demographic information in addition to the dependent measures. After the interactions, participants can provide optional feedback for the study.

4.3 Analysis

We employ a mixed-methods approach combining conversational analysis of human-AI dialogues, quantitative analysis of the measured variables, and qualitative analysis of behavioral dimensions associated with successful coaching sessions.

4.3.1 Conversational Analysis. We conduct descriptive analyses of three key variables: session length, MI strategies employed by AI coaches, and empathy levels expressed by AI coaches.

Annotating MI strategies: The coaching agents may fail to demonstrate behaviours consistent with MI principles. All conversations are coded to assess MI consistency and compare strategy usage between agents. We utilise `GPT-4o` to obtain results on MI strategy usage according to the Motivational Interviewing Treatment Integrity (MITI) guideline [48]. Researchers have suggested GPT models can serve as excellent data annotators, potentially replacing crowdsourced workers [16, 25, 32, 34]. Jörke et al. [35] found that

their LLM-based coding results are highly analogous to their expert coding results, revealing similar patterns in MI strategy usage.

The annotation process consists of 11 MI strategies classified into three groups: 1 neutral strategy (Others), 2 MI-non-adherent strategies (Persuade without Permission, Confront), and 8 MI-adherent strategies (Give Information, Persuade with Permission, Open Question, Closed Question, Reflection, Emphasize Autonomy, Affirm, Seek Collaboration). We adapted the prompt by Jörke et al. [35]. Each strategy in the prompt includes one positive example and two negative examples, all drawn from the MITI guidelines. For each API call, GPT-4o receives a three-turn conversational snapshot and annotates the final AI coach utterance. The complete prompt is provided in Table 11 in Appendix C.

Measuring empathy levels: Given empathy's central role in MI, we measure empathy levels from both coaching agents to determine whether they are appropriately demonstrated. To quantify empathy expressed by the coaching agents in the conversations, we employ the `EPITOME` framework [63]. We train the model using the authors' open-source codes and their dataset of 10k Reddit posts. `EPITOME` evaluates empathy across three distinct dimensions, including Emotional Reactions, Interpretations, Explorations. For each user-agent interaction pair, the model assigns scores of 0 (no signal), 1 (weak signal), or 2 (strong signal) across all three dimensions, yielding a maximum total empathy score of 6 per response.

4.3.2 Analysis of User Perception Measures. We first report summary statistics for all dependent measures for both agent settings, namely ΔReadiness, user perception of MI (CEMI), and therapeutic rapport (WAI) after interacting with the coaching agent. For our main quantitative analysis, we address the following hypotheses:

- **H1** Compared to those with the base-MI systems, users interacting with the MI-aware agent will show greater increases in motivation for change.
- **H2** Compared to those with base-MI systems, users interacting with the MI-aware agent will rate its MI capabilities more favourably.
- **H3** Compared to those with base-MI systems, users interacting with the MI-aware agent will report higher levels of therapeutic rapport.

Each hypothesis is evaluated using one-tailed tests, either independent parametric t-tests or non-parametric alternatives based on whether the data meets normality and homogeneity assumptions. The independent variable is the type of coaching agent (MI-aware or base-MI) and the dependent variables are (1) therapeutic alliance (WAI), (2) perception of MI (CEMI), and (3) the change in readiness to change from pre- to post-interaction (ΔReadiness).

We also examine correlations among these dependent variables. Given that therapeutic alliance is considered the strongest predictor of coaching outcomes, we expect ΔReadiness and WAI to be positively correlated.

4.3.3 Analysis on Factors Contributing to Coaching Success. Beyond condition-specific elements, common factors inherent to coaching interactions likely influence participants' readiness to change. To identify these factors, we conduct qualitative and exploratory analyses examining the coaching behaviours associated with successful coaching sessions. Successful sessions are defined as sessions in which users' motivation for change increases after the interaction. We first perform qualitative coding on the human-agent.

Table 3: Counts and percentages of three behavioural dimensions per agent condition.

Condition	Plan Agreement	Proactivity	Cooperation
MI-aware	42 (60.0%)	39 (55.7%)	51 (72.8%)
base-MI	44 (62.8%)	39 (55.7%)	53 (75.7%)

Coding Behavioural Dimensions: To identify which specific elements of human-agent interaction contribute to these positive attitudinal outcomes, we conduct a deductive thematic analysis [9] using therapeutic alliance theory [8] as our conceptual framework. We utilise "Goal/Task" and "Bond" as two main themes, merging "Goal" and "Task" into one dimension as studies showed that clients face difficulty distinguishing these two dimensions when rating [24]. In an initial coding phase, the first author familiarised themselves with the WAI subscale definitions and read through all collected conversations to identify emerging sub-codes based on these definitions. Then the sub-codes are sorted and merged, yielding three dimensions that we believe best represent the behaviours of users and AI coaches emerging from the data. The coding scheme is as follows:

- Plan Agreement ("Goal/Task" Theme): The coaching agents collaborate with users to develop specific, concrete change plans by the end of each session. Specific plans include information such as the agreed-upon activities, the timing/frequency, and the duration. Conversely, sessions that fail to reach agreement on plans occur when users do not explicitly agree to proposed actions or when the plan lacks specificity.
- Proactivity ("Goal/Task" Theme): Proactive users engage in developing their exercise plans. Rather than seeking quick fixes or expecting ready-made solutions, they work collaboratively with the AI coaches. Non-proactive users, in contrast, rely heavily on the AI coaches for direction, frequently asking for suggestions and recommendations while contributing minimal input themselves.
- Cooperation ("Bond" Theme): Cooperative users consistently reply in full sentences and makes an effort to respond to the AI coaches, answering questions thoughtfully and contributing meaningfully to the conversations. In contrast, uncooperative users tend to provide brief, minimal responses with little effort to engage meaningfully.

"Plan Agreement" is selected based on the definition of the "Goal/Task" scale, which measures coaches' and clients' mutual agreement on goals and tasks. "Proactivity" was chosen because prior research demonstrates that client-initiated action plans positively correlate with coaching success [24]. For "Cooperation", we draw on the "Bond" scale, which captures the emotional connection, trust, and confidence between coaches and clients. We interpret users' level of engagement and cooperation as indicators of their trust in the agents' guidance capabilities. These two user behaviours, namely cooperation and proactivity, align with findings from Meyer and Elsweiler [46], who identified similar

Table 4: The number of conversations per readiness change direction (i.e., increased, unchanged, and decreased) in the presence and absence of each behaviour.

Behaviour		Inc.	Unch.	Dec.	Total
Plan Agreement	yes	56	28	2	86
	no	26	22	6	54
Proactivity	yes	50	25	3	78
	no	32	25	5	62
Cooperation	yes	67	31	6	104
	no	15	19	2	36

patterns through inductive analysis in their study of MI conversational agents for behaviour change support. Further details on the annotation guidelines and example chats are in Appendix B.

The first author conducted binary annotations of all human-AI interactions, coding each dimension as either present or absent. We then perform comparative analyses to examine how agent and user behaviours influence interaction outcomes. The results offer insights into human-AI interaction dynamics within physical activity promotion more broadly and thus can inform the design of future behaviour change systems.

Performing Analyses: Table 3 shows the distribution of three behavioural dimensions per agent condition. We conduct chi-square tests to examine whether each behavioral dimension is associated with study condition. None of the tests yields significant results, indicating that the distributions of these dimensions do not differ significantly across study conditions. While our user study distinguishes between base-MI and MI-aware conditions to test which agent leads to better outcomes, we now focus on identifying which elements of coaching sessions contribute to success regardless of condition. We therefore collapse across study conditions to analyse behavioural factors contributing to coaching success.

Table 4 presents the distribution of change directions in the presence and absence of each behaviour. We examine whether the presence of a behaviour increases ΔReadiness following interaction with the agents. To test this, we conduct one-tailed Mann-Whitney U tests with ΔReadiness as the dependent variable and three behaviours as independent variables. We form further three following hypotheses:

- **H4** The presence of "Plan Agreement" behaviour will lead to greater increases in motivation for change.
- **H5** The presence of "Proactivity" behaviour will lead to greater increases in motivation for change.
- **H6** The presence of "Cooperation" behaviour will lead to greater increases in motivation for change.

Additionally, in MI, where counsellors facilitate clients' intrinsic motivation for change rather than prescribe solutions, empathy is a foundational ingredient and serves as a key indicator of therapist effectiveness [78]. Given empathy's crucial role in establishing rapport with coaching agents, we investigate its correlation with users' readiness to change. We compute Pearson correlation coefficients between session-wide `EPITOME` scores (Section 5.3) and ΔReadiness.

We also analyse users' feedback to gain insights into their experiences interacting with the AI coaches.

4.4 Participants

We use G*Power[9] to calculate appropriate sample size with the following parameters: one-tailed t-test, a medium effect size of d = 0.5, a power of $\beta = 0.8$. The effect size is based on Singh et al. [66], whose systematic review reported small to medium effect sizes for behaviour change outcomes in chatbot interventions. In the main analysis (Section 4.3.2), we apply Bonferroni correction to account for multiple hypothesis testing across three hypotheses, setting the significance threshold at $\alpha = 0.05/3 = 0.0167$. The analysis indicates a required sample size of 72 participants per agent condition. With two agents in our study, the total target sample size is 144 participants.

We recruit participants through Prolific channel[10]. Eligible criteria include (1) age between 18 and 69 years, (2) fluency in English, (3) engagement in physical exercise for fewer than 150 minutes per week. We further screen out Prolific participants and only accept those (4) who were born and are currently living in English-speaking countries, and (5) having had a minimum of 50 previous submissions with at least a 95% approval rating. Compensation is aligned with Prolific's recommended minimum rate (£9.00 per hour). Given the expected 30-minute duration, participants receive a flat rate of £5.00 for joining the study. Ethics application for the study has been approved by the Research Ethics Committee of Technological University Dublin with reference number being REC-24-022.

Each Prolific submission is manually reviewed to ensure participants provide all required data. We exclude 4 of 144 submissions: 3 due to suspected AI usage and 1 due to low interaction quality (random responses and premature chat termination). This yields 70 data samples per agent condition.

Participant demographics are listed in Table 5. Participant ages are similarly distributed across groups. The majority of participants are white and currently residing in the US and UK. Three participants self-identified as having a disability during the interaction with the AI coaches. The majority have previously interacted with AI chatbots with one participant reported being an AI developer.

Figure 3 shows the frequency count of the user readiness to change, indicated by the Contemplation Ladder, before interacting the coaching agents. The distribution of the MI-aware setting shows greater balance compared to the base-MI setting, which exhibits a right-skewed distribution. The readiness level 8 ("I am starting to think about how to be more physically active") receives the highest count in both settings. According to the transtheoretical model [58], users are mostly either in the contemplation or preparation stages. Stage classification follows the mapping from contemplation ladder scores [7] to stages of change by Slavet et al. [67]: precontemplation (0-3), contemplation (4-6), preparation (7-8), action (9), and maintenance (10).

[9]https://www.psychologie.hhu.de/arbeitsgruppen/allgemeine-psychologie-und-arbeitspsychologie/gpower
[10]https://www.prolific.com/

Table 5: Summary of participant demographic information in our user study ($N = 140$).

Age Group	18-25: 18, 26-30: 23, 31-35: 19, 36-40: 19, 41-45: 14, 46-50: 17, 51-55: 15, 55+: 15
Gender	Male: 75, Female: 63, Non-binary: 2
Ethnicity	White: 106, Asian: 6, Black: 15, Mixed: 12, Other: 1
Residence	United States: 65, United Kingdom: 66, Canada: 9
Education	High School: 47, Bachelor's: 61, Master's: 18, PhD: 5, Other: 9
Stage of Change	Precontemplation: 18, Contemplation: 45, Preparation: 60, Action: 8, Maintenance: 9
AI Experience	Very Little: 38, Some: 54, A lot: 47, Expert: 1

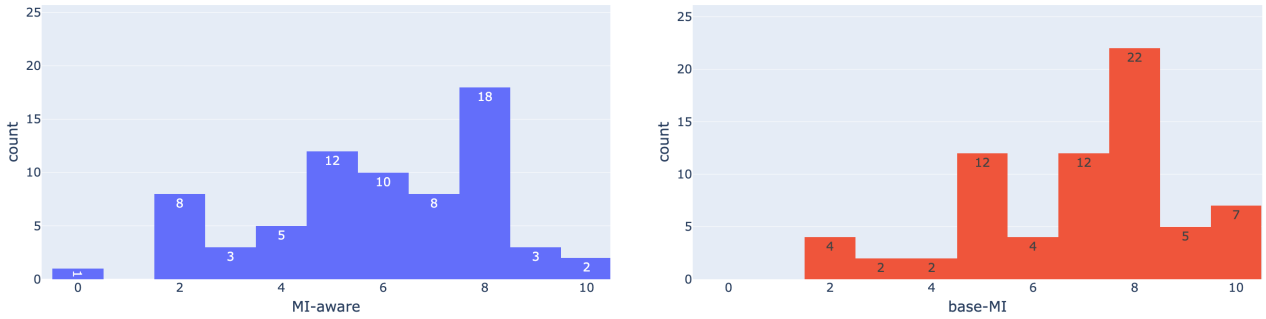


Figure 3: Distribution of user readiness to change before interacting with the coaching agents for both agent conditions.

5 Conversational Analysis

5.1 Response Length: How long are the users' responses to the systems?

Table 6 presents conversation length statistics for both agent settings. The number of turns is comparable across conditions (M = 17, range = 8-21). While turn counts are similar, participants in the MI-aware setting engage more with the agent, averaging 236 words per chat compared to 179 words in the base-MI condition–a 32% increase in text output. Mann-Whitney U tests confirm statistically significant differences between conditions for both total word counts ($U = 3011.5, p = 0.019$) and word/turn ratios ($U = 2970.5, p = 0.03$).

5.2 MI Strategies: How do the agents employ MI strategies?

Table 7 presents the distribution of MI strategies employed by both systems. "Questions" emerged as the most frequently used strategy (39.66% for the MI-aware agent and 43.86% for the base-MI agent), followed by "Reflections" for the MI-aware agent (13.79%) and "Persuade with Permission" for the base-MI agent (14.43%). Among strategies designed to provide emotional and motivational support, namely "Reflections", "Affirm", and "Emphasise Autonomy", the base-MI system favours the "Affirm" strategy, while the MI-aware system demonstrates a preference for "Reflections".

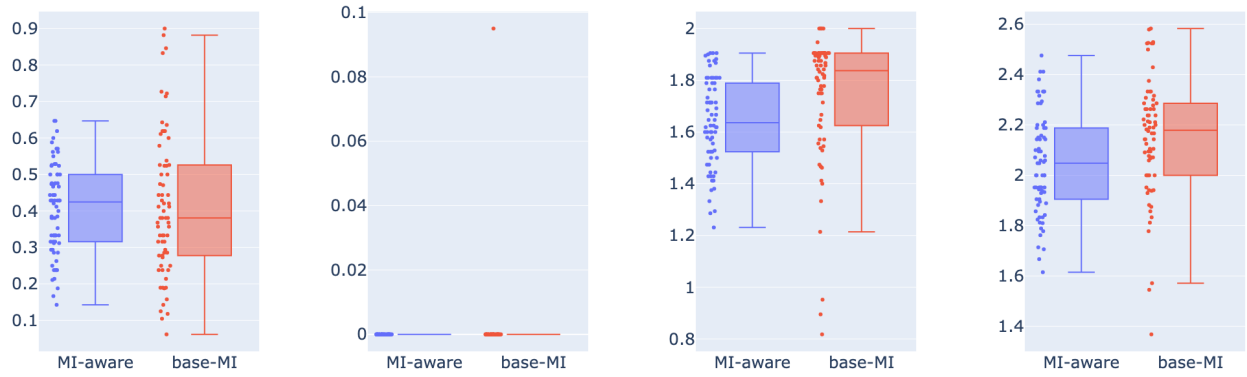
MI guidelines [47, 48] recommend that when employing "Persuade" and "Give Information" strategies, practitioners should also

Table 6: Mean and standard deviations of conversation lengths for both settings.

Setting	Length Value	Agent	Human
MI-aware	Word	556.1 (± 127.1)	236.1 (± 162.6)
	Turn	17.8 (± 3.4)	17.8 (± 3.4)
	Word/Turn	30.5 (± 2.3)	12.9 (± 8.6)
base-MI	Word	441.7 (± 123.9)	178.9 (± 102.2)
	Turn	17.3 (± 3.7)	17.3 (± 3.7)
	Word/Turn	24.8 (± 2.6)	9.6 (± 4.7)

Table 7: Mean and standard deviations of MI strategy usage per chat by both agents.

Strategy	MI-aware	base-MI
Questions	39.66 (± 7.83)	43.86 (± 12.34)
Reflections	13.79 (± 5.44)	6.10 (± 4.73)
Affirm	5.37 (± 3.03)	11.15 (± 8.31)
Seek Collaboration	9.31 (± 4.90)	12.39 (± 8.36)
Emphasise Autonomy	8.68 (± 3.61)	1.68 (± 2.56)
Give Information	5.82 (± 4.01)	4.70 (± 6.70)
Persuade with Permission	10.88 (± 5.24)	14.43 (± 8.12)
Persuade without Permission	0.25 (± 0.85)	0.96 (± 1.84)
Others	6.23 (± 2.16)	4.72 (± 4.46)



(a) Emotional Reactions **(b) Interpretations** **(c) Explorations** **(d) Total Empathy**

Figure 4: Box plots display session-wide empathy levels per agent condition.

use "Seek Collaboration" to request permission or "Emphasise Autonomy" to acknowledge clients' freedom of choice. Analysis of strategy usage reveals that both systems adhere to this principle, with compliance rates of 72% for the MI-aware system and 50% for the base-MI system.

Regarding unsolicited advice ("Persuade without Permission"), a pattern associated with low-quality human therapy conversations [12, 35], we find occurrence rates of 1% for the base-MI system and 0.25% for the MI-aware system. However, the base-MI may initiate persuasive behaviours as early as turn 1. The MI-aware agent is designed to defer such behaviours until turn 8, though they occasionally occur as early as turn 5. Although restricting systems from offering advice too early in conversations presents challenges, we observe that this issue stems primarily from users' tendency to request advice or quick fixes very early in the interaction. As a result, LLMs tend to comply with users' requests in their responses due to their training regime in Reinforcement Learning from Human Feedback, which biases them toward single-turn question answering.

5.3 Empathy Levels: How empathic are the agents' responses to users' texts?

Figure 4 presents the box plots displaying the session-wide empathy levels. These scores are computed by averaging all scored interaction pairs within a session, an approach found to correlate with overall session empathy levels [73]. Both systems yield averaged total empathy scores between 1.3 and 2.1 out of 6. Although these scores appear relatively low, they exceed the averaged score of 1.09 reported in the original EPITOME study using Reddit posts [63] and align with findings from [73]. In that study, averaged empathy scores ranged from 0.8 to 2.7, yet 85% of users reported feeling *"heard and understood"*.

Generic GPT-4 achieve high scores in Emotional Reactions but low scores in Interpretations and Explorations [13]. In contrast, both the base-MI and MI-aware agents demonstrate the inverse pattern: low scores in Emotional Reactions and Interpretations but high scores in Explorations. This suggests that rather than responding emotionally to user utterances, both agents adopt an exploratory approach, probing users' experiences through targeted questioning.

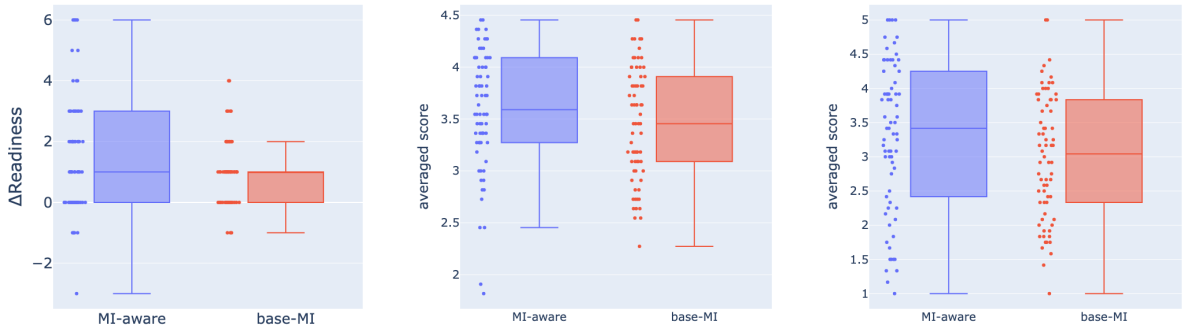
Interpretations scores are nearly absent in both conditions, which can be explained by the model's documented weak performance on this dimension [64, 73] and the agents' lack of voicing their own experiences (e.g., "I also have..."). For Emotional Reactions, both conditions show similarly low mean scores (0.40 for MI-aware, 0.41 for base-MI), though the base-MI condition exhibits greater variance. Conversely, both conditions score highly on the Explorations dimension, with scores of 1.64 for MI-aware and 1.74 for base-MI agents. Responses containing questions are found to yield high scores on the Exploration dimension [63, 73]. Since the base-MI agent employs the "Questions" strategy more frequently than the MI-aware agent (See Table 7), its higher Explorations score is expected. However, higher scores on this dimension do not necessarily indicate superior performance, as Reddit responses with strong exploration received 44% fewer likes [63], and excessive use of open-ended questions can confuse users and lead to lower rating of the sessions [73].

6 Analysis of User Perception Measures

Figure 5 presents the box plots and Table 8 presents the summary statistics of all dependent measures for both settings, namely ΔReadiness, user perception of MI (CEMI), and therapeutic rapport (WAI) after interacting with the AI coach.

6.1 Readiness to Change: How ready are the users after the interaction?

6.1.1 Summary Statistics: ΔReadiness measures the difference in user readiness to change post- and pre-experiment. Table 9 shows the distribution of readiness change directions across study conditions. Across all 140 conversations, the majority (58.57%) resulted in increased readiness to change, while readiness remained unchanged in 35.71% of cases. However, 4.7% of conversations led



(a) ΔReadiness (b) CEMI scale (c) WAI scale

Figure 5: Box plots for results of all measures for both settings: (a) difference in readiness to change pre- and post-intervention, (b) user perception of MI and (c) therapeutic rapport after interacting with the coaching agents.

Table 8: Means, standard deviations, 95% confidence intervals, and Cohen's d effect size for all measures for both settings.

Measure	MI-aware			base-MI			Cohen's d
	Mean	SD	CI	Mean	SD	CI	
ΔReadiness	1.47	1.87	(1.02, 1.92)	0.84	1.08	(0.58, 1.10)	0.41
CEMI	3.59	0.57	(3.45, 3.73)	3.46	0.54	(3.33, 3.59)	0.23
WAI	3.30	1.10	(3.03, 3.56)	3.00	0.92	(2.78, 3.22)	0.29

Table 9: The number of conversations for each readiness change direction after the interaction per study condition.

Condition	Increased	Unchanged	Decreased	Total
MI-aware	43	22	5	70
base-MI	39	28	3	70
Total	82	50	8	140

to decreased user readiness. As shown in Figure 5a, the MI-aware system demonstrates greater variance in outcomes compared to the base-MI system. The MI-aware system produces both the largest increases in readiness ($\Delta > 4$) and the most significant decrease ($\Delta = -3$). Notably, all users experiencing substantial improvements ($\Delta \geq 3$) are in the precontemplation or contemplation stages, while those showing decreased readiness are more likely in the preparation or maintenance stages. This pattern reinforces findings that MI is most effective for clients with low motivation [20].

6.1.2 Hypothesis Testing: We examine between-group differences in Readiness to Change scores across conditions and analyse within-group repeated measures to determine if chatbot interactions affected participants' readiness to change.

Attitudinal Changes Between Groups: Due to violations of both normality and homogeneity of variance assumptions, we employ the Mann-Whitney U test to compare effectiveness between groups. There is no statistically significant difference between the MI-aware and base-MI groups ($p = 0.034 > 0.0167$). Thus, H1 is not supported by the data.

Attitudinal Changes Within Each Group: As the normality assumption was violated, we employ the Wilcoxon signed-rank test to assess intervention effects within each group. Both the MI-aware and base-MI groups demonstrate highly significant changes from pre- to post-intervention (MI-aware: $W = 70.5, p = 8.6e-08$; base-MI: $W = 42, p = 1.2e-07$).

This motivational increase is more pronounced among participants in earlier stages of change. After mapping 10-point contemplation scores to the five stages of change [58], Spearman's correlation analysis revealed a moderate negative correlation ($\rho = -0.52$, $p = 4.1e-11$) between pre-experimental stage of change and magnitude of improvement. This indicates that individuals at earlier stages of change exhibit greater improvements in readiness scores post-experiment, while those at later stages show less dramatic gains.

6.2 Perception on MI (CEMI)

6.2.1 Summary Statistics: For both CEMI and WAI measures, the score ranges from 1 to 5 with higher scores indicating more positive user experiences across all aspects. For analysis purposes, we calculate composite scores by averaging responses across all items. Note that scores for relational items in the CEMI survey have been reverse-coded. Figure 5b reveals more similar distributions between the two systems compared to the stark differences observed in ΔReadiness (Figure 5a). The CEMI measure displays

more tightly clustered score distributions for both conditions, with reduced extreme variability compared to WAI (Figure 5c).

6.2.2 Hypothesis Testing: The hypothesis that participants would rate the MI-aware agent more favourably than the base-MI agent on Perception of MI (H2) is not supported by the data. Mann-Whitney U tests reveals that there is no statistically significant difference between the MI-aware the base-MI agents ($p = 0.052 > 0.0167$). Further analysis indicates no significant differences between groups for relational items; however, significant differences emerge for technical items ($U = 3044.5, p = 0.0065$), implying that participants perceived the MI-aware agent as significant more MI-like than the base-MI agent.

6.3 Therapeutic Alliance (WAI)

6.3.1 Summary Statistics: Unlike the stark differences observed in ΔReadiness (Figure 5a), the distributions of WAI (Figure 5c) and CEMI (Figure 5b) are more similar to each other. For the WAI measure, the MI-aware condition exhibits greater variability with scores spanning from approximately 1.0 to 5.0, while the base-MI condition shows a more compressed distribution. Despite these similarities in distribution patterns, the MI-aware condition shows a trend towards higher mean scores on both measures: CEMI (3.59 vs. 3.46) and WAI (3.30 vs. 3.00) compared to the base-MI condition.

6.3.2 Hypothesis Testing: The means presented in Table 8 and the boxplots in Figure 5 indicate stronger therapeutic alliance values for the MI-aware agent. However, the data do not support H3, which predicts that participants would rate the MI-aware agent higher than the base-MI agent on therapeutic alliance ($p = 0.035 > 0.0167$). Further analysis reveals that among all subscales (Goal, Task, Bond), only the task-related items show a marginally significant difference between two groups ($p = 0.025$).

6.4 Correlation: How do the dependent variables correlate with each other?

Figure 6 presents correlation matrix between dependent variables. CEMI and WAI exhibit a strong positive correlation (Pearson coefficient $r = 0.71$). However, similar to Meyer and Elsweiler [46], we observe no significant correlation between either measure and ΔReadiness, suggesting that subjective user experiences of the conversations do not necessarily correspond to intervention effectiveness. ΔReadiness correlates only with users' initial readiness scores before interacting with the coaching agent ($r = -0.59$). This negative coefficient indicates that users with higher baseline readiness to change experience smaller improvements following the conversations.

7 Analysis on Factors Contributing to Coaching Success

Successful sessions are defined as sessions in which users' motivation for change increases after the interaction. We perform analyses on coaching behaviours, empathy levels, and user's feedback.

7.1 Effects of Behaviours On Session Outcomes

We report our findings on the relationship between successful coaching sessions with three behavioural dimensions derived from our qualitative coding based on the therapeutic alliance theory. We find significant effects for "Plan Agreement" ($p = 0.038$) and marginally significant effects for "Cooperation" ($p = 0.044$), but no significant effect for "Proactivity" $p = 0.052$). These results support H4 and H6 but reject H6. This indicates that users who reach agreement with agents on a behaviour change plan are most likely to experience greater increases in their intrinsic motivation to change.

7.2 Effects of Empathy Levels on Session Outcomes

All empathy sub-scores, namely emotional reactions, interpretations, explorations, and total empathy, show weak correlations with users' readiness to change ($r < 0.2$). While Syed et al. [73] found a negative correlation between exploration scores and user ratings ($r = -0.565$), we observed no such negative correlation between exploration scores and either users' motivational outcomes or their session experiences, despite our averaged exploration scores being higher than theirs.

7.3 Insights from Users' Feedback

Upon completing the study, users can provide optional feedback about the coaching agents. Most users choose to leave their feedback, and we analyse these responses across all study conditions to gain insights into user experiences.

7.3.1 Rewarding experiences among users: We find that participants overall had positive experiences interacting with the coaching agents, particularly praising their supportive and non-judgmental language. For example, User 120 (MI-aware condition) shared, *"They were very proactive and communicated respectfully with me about my goals."* while User 96 (MI-aware condition) noted that *"Jordan was polite and I feel like he wasn't judgemental."*.

Many participants were impressed by the agents' ability to motivate them toward physical activity. User 52 (base-MI condition) remarked, *"Actually could convince me to exercise!! Very good AI,"* and User 28 (MI-aware condition) observed, *"The session was more helpful than I expected, I could use an AI coach for other aspects of my life now."*. User 37 (MI-aware condition) particularly valued the agent' non-prescriptive approach and thoughtful questioning, appreciating how the agent *"...made me think as to how i can change my behaviour for the good, it really felt as if i was talking to an intelligent human."*

7.3.2 Solution-oriented expectations among users: Instead of the the agents' facilitative, non-prescriptive approach, many participants express preferences for more direct, prescriptive guidance. User 59 (base-MI condition) shared, *"The AI coach didn't give me as much advice as I expected."* while User 121 (MI-aware condition) noted, *"The AI chat bot does not go extensively into advising me. In needs to go deeper like an expert would do. "*

The non-prescriptive principles of MI assumes users can come up with optimal solutions through guided self-discovery using open questions, reflections, and affirmations. Yet, participants expressed their desire for the AI systems to go beyond this approach and provide feedback on their choices. For example, User 48 (base-MI condition) felt, *"it was (only) mirroring what i already said"* while User 93 (base-MI condition) *"Jordan needs to actually give advice,*

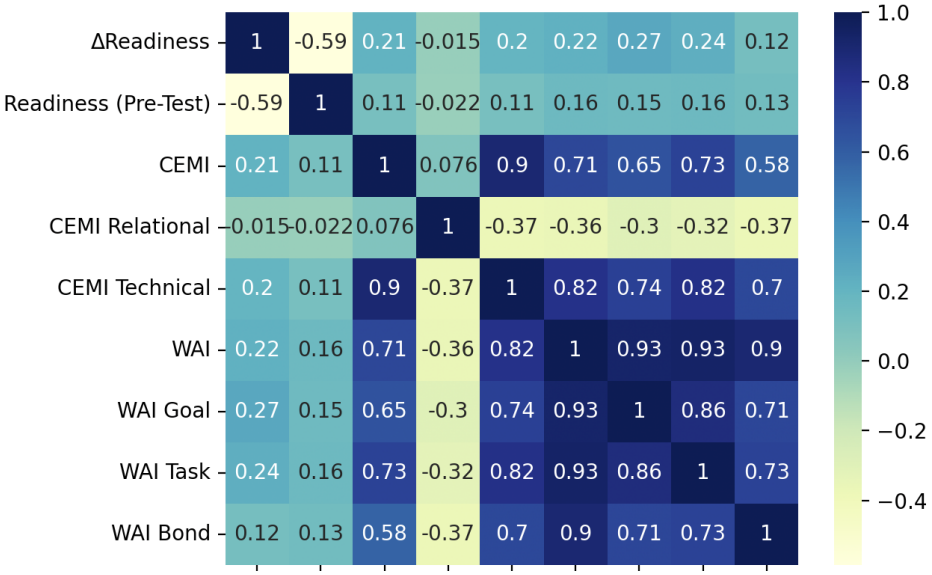


Figure 6: Pearson correlation matrix of pre- and post-study measures.

not just act positive about what I'm already planning.". Similarly, User 109 (MI-aware condition) noted, *"the AI coach tried to get me to come up with my own solutions rather than helping to guide me or make suggestions."*

Additionally, some participants felt overwhelmed by the agents asking too many questions, which they perceived as lacking adequate guidance User 82 (base-MI condition) observed, *"Everything was question based. Made it feel very one sided."* while User 40 (MI-aware condition) noted, *"it didn't really give advice, was just constant questions without much feedback".* This pattern likely stems from GPT-4o's tendency to generate multiple questions within a single response.

8 Discussion

In this study, we present a coaching framework that can select appropriate coaching strategies by modelling user motivational states. Here, we first summarise our key quantitative and qualitative findings, and reflect on how they align with expert opinions that inform the design principles of our MI-aware agent. Finally, we discuss the implications for LLM-based health coaching and suggest future design considerations.

8.1 Digital Health Coaching to Promote Physical Activity

The transtheoretical model by Prochaska and Velicer [58] proposes that users at different stages of change require tailored strategies. This paper introduces an MI-aware coaching framework that uses user motivational states to select appropriate coaching strategies aligned with their current stage of change. We tested the effectiveness of our framework when delivered through a chatbot to promote physical activity by comparing it against a baseline system that is simply instructed to act as a coach using MI techniques without any stage-specific prompt engineering. No significant differences emerged between conditions, consistent with He et al. [31], Meyer and Elsweiler [46], and Geng et al. [23], who found no effect of MI-based interventions compared to baselines on behaviour-related outcomes (quit intentions, readiness to change, and perceived premenstrual syndrome). However, both conditions significantly increased readiness to change, suggesting brief motivational coaching chatbot interactions can promote behavioral change intentions. One possible explanation is that users may value the journey of exploration and framing more than the goal itself [69].

Exploratory analysis of CEMI and WAI subscales showed no significant differences between coaching agents on relational or bonding items. However, participants rated the MI-aware agent as significantly more adherent to technical MI skills (as measured by CEMI technical items) and more effective at exercise planning (as measured by WAI task-related items) than the base-MI agent.

Qualitative data further expanded the experimental findings. Although our between-group analyses did not support the main hypotheses, within-group comparisons revealed that both MI-aware and base-MI conditions significantly increased participants' readiness to change. This motivates us to identify common factors inherent to the coaching interactions that drove these behavioral shifts. The behaviours we identified using therapeutic alliance theory as our conceptual framework indicate that users who reach agreement with agents on a change plan and those who behave more cooperatively are more likely to experience greater increases in their intrinsic motivation to change. Our findings partially align with those of Meyer and Elsweiler [46], who found no significant effects for either "Cooperation" or "Proactivity". However, their coaching agent differed from ours in that it did not assist users in planning their change actions but instead asked users to come up with a change plan for themselves. Therefore, the borderline significant effect we observe for "Cooperation" may stem from differences in either data samples or system design.

In our data, all empathy sub-scores demonstrated weak correlations with participants' readiness to change. Contrary to findings by Syed et al. [73], we observed no negative correlation between exploration scores and either participants' motivational outcomes or their session experiences. This discrepancy may stem from contextual differences: their study involved emotional support using Cognitive Behaviour Therapy, where excessive questioning could make users feel unheard or misunderstood. In contrast, we study MI coaching systems for behaviour change, which is inherently more solution-oriented. During the planning phase, questions about users' preferred activities, and choices are contextually appropriate and overall do not cause the same negative reactions.

8.2 Balancing Between Coaching Principles and Users Expectations

Our qualitative findings reveal a tension between user preferences and MI coaching principles regarding AI coaching behaviours. As a client-centred, non-prescriptive approach, MI emphasises emotional and motivational support while encouraging clients to develop their own solutions rather than providing direct answers. MI experts discourage problem-solving and advice-giving behaviours, as ready-made solutions don't facilitate lasting change. As such, following MI principles, the MI-aware agent solicits users' own plans before offering guidance, and provides options only when requested.

However, our participants demonstrated different expectations, often being solution-oriented from the start. Despite the agent's explicit statement that it would explore their motivation first before planning actions, several participants persisted in requesting direct solutions. User 68 even expressed frustration during the interaction, demanding: *"Are you going to offer me some specific recommendations here or are we going to just discuss meaningless feelings?"* Additionally, MI agents are designed to accept clients' ideas without challenge, operating under the assumption that users possess the best understanding of their own situations. However, participants wanted AI systems to elaborate on and provide feedback on their choices to develop personalised plans. This suggests a strong user preference for prescriptive guidance that directly contradicts core MI methodology. In a similar user study, Srinivas et al. [69] found that although some users appreciated the agent's goal-seeking behaviour, they consistently rated it low on overall satisfaction, suggesting a negative correlation between motivation seeking and user satisfaction.

The advice-focused mindset among participants is likely attributable to the fact that more than half were in later stages of change (see Table 5). The transtheoretical model [58] hypothesises that such individuals have already recognised the need for change and consequently expect more directive guidance. In contrast, those in earlier stages of change remain ambivalent about change, and are therefore more willing to explore their motivation.

Our coaching agent design allocates six conversational turns to assess users' initial motivation levels, then adapts its approach to enhance motivation or set change goals as the conversation evolves. Future research could explore directly obtaining this information from users and leveraging it to direct them toward suitable coaching systems: whether more supportive or advice-oriented approaches.

8.3 Prioritising Core Tasks Over Bonding Skills

In the qualitative analysis, we identified elements that contribute to successful coaching sessions. Research in human coaching demonstrates that the therapeutic relationship is the strongest predictor of outcomes [27, 75]. Using this as our conceptual framework, we found that "Plan Agreement", which is the behaviour aligned with the "Goals/Tasks" subscale, was most strongly associated with increased attitudinal outcomes. This finding aligns with human coaching research showing that the "Goals/Tasks" subscale is a strong predictor of coaching success, while the "Bond" subscale does not significantly influence outcomes [2, 24, 26, 68].

Experts have expressed skepticism about AI agents' ability to establish genuine bonds with humans [57]. This was reflected in our study, where User 12 stated they assigned low scores across multiple items because they knew they were interacting with an AI that lacks genuine emotions and feelings. Given this challenge and the importance of goals and tasks in the coaching process, coaching systems designed to support behaviour change might benefit more from focusing on planning change behaviours rather than maximising human-like empathy. The coaching system developed by Duggan et al. [17] lacked human-like features and did not provide empathy, focusing instead on meaningful goals and activities. Despite this, users reported increased awareness and motivation, and the system simulated key aspects of conventional human coaching interactions. This suggests that personalised tasks and goals can support relationship formation, with user trust and confidence in their system emerging from its effective goal-setting capabilities. Research by Gómez Penedo et al. [29] further supported the hypothesis that in digital interventions, patient alignment with therapeutic tasks and goals may be more critical than forming therapeutic bonds with supporters. Consistent with prior findings, Srinivas et al. [69] found that in AI coaching contexts, stylistic elements such as motivational tone and encouragement are perceived negatively when core functional components are absent. This suggests that effective style enhances user experience only when foundational task-related components are present.

We do not imply that fostering bonds between users and coaching agents is unnecessary or unimportant. Expressions of empathy can enhance perceived warmth, and build trust [56, 62]. However, empathy can be counterproductive. Seitz [62] discovered that empathy delivered in an empathetic[11] or sympathetic[12] manner may reduce chatbot authenticity and undermine trust. Additionally, excessive empathy risks reinforcing harmful behaviours and perspectives [13]. More importantly, it may also prevent growth for individuals who need discomfort to motivate change and self-improvement [13, 59]. In some cases, articulating negative experiences while receiving empathy and validation could intensify users' perceived distress [23]. Therefore, future behaviour change systems should prioritise goal-oriented interactions rather than maximising human-like empathy.

9 Limitations

We acknowledge several limitations in our study. First, participants were recruited from Prolific and may lack genuine motivation for

[11] e.g., "I can well understand your concerns/situation."
[12] e.g., "I feel sorry for you."

behaviour change, potentially affecting the authenticity of their engagement. Second, our system was designed for a single human-agent interaction session, whereas meaningful behavioral change typically requires sustained, multi-year engagement. However, we believe that capturing user preferences in initial interactions remains valuable, as positive first experiences are crucial for fostering continued participation in future sessions. Third, each phase has a fixed number of conversational turns, which may affect user experience. We found that this constraint worked well for most participants, though some found the sessions either too long or too short. One solution would be to implement additional modules that monitor whether core objectives have been met and signal when to conclude each phase. However, this approach could increase inference time and delay agent responses, negatively impacting user experience. For this reason, we opted for a fixed number of turns. We anticipate that advances in LLM inference speed will eventually address this limitation. Finally, our system relies exclusively on textual features to detect motivational states, ignoring non-verbal cues like facial expressions that can signal resistance such as silence or averted gaze. Future iterations should therefore explore multi-modal inputs, such as speech and visual data, to create a more comprehensive and natural user experience.

10 Conclusion

In this paper, we present a coaching framework that enables an agent to dynamically adapt MI strategies based on clients' stages of change through motivational language detection. Our framework provides a transparent, controllable, and scalable approach to digital interventions, with potential for cross-domain adaptation through prompt modification and domain-specific knowledge integration. We evaluated the system through both quantitative and qualitative analyses in a user study with 140 participants. Results demonstrate significant increases in users' readiness to change following interactions with motivational coaching chatbots. Qualitative analysis further reveals that users who reach agreement with agents on a change plan before session completion experience the greatest increases in intrinsic motivation. These findings inform the design of AI health coaches for behaviour change support tailored to diverse user groups.

Acknowledgments

This publication has emanated from research conducted with the financial support of Science Foundation Ireland under Grant number 18/CRT/6183 (ML-Labs) and the ADAPT SFI Research Centre for AI-Driven Digital Content Technology under Grant No. 13/RC/2106_P2.

References

[1] Abubakar Abid, Ali Abdalla, Ali Abid, Dawood Khan, Abdulrahman Alfozan, and James Y. Zou. 2019. Gradio: Hassle-Free Sharing and Testing of ML Models in the Wild. https://arxiv.org/abs/1906.02569 Presented at 2019 ICML Workshop on Human in the Loop Learning (HILL 2019).

[2] Faris Alodaibi, Jason Beneciuk, Rett Holmes, Stephen Kareha, Deanna Hayes, and Julie Fritz. 2021. The Relationship of the Therapeutic Alliance to Patient Characteristics and Functional Outcome During an Episode of Physical Therapy Care for Patients With Low Back Pain: An Observational Study. *Physical Therapy* 101, 4 (April 2021), pzab026. doi:10.1093/ptj/pzab026

[3] Paul C Amrhein, William R. Miller, Theresa Moyers, and Denise Ernst. 2008. Motivational Interviewing Skill Code (MISC) 2.1.

[4] Paul C. Amrhein, William R. Miller, Carolina E. Yahne, Michael Palmer, and Laura Fulcher. 2003. Client commitment language during motivational interviewing predicts drug use outcomes. *Journal of Consulting and Clinical Psychology* 71, 5 (Oct. 2003), 862–878. doi:10.1037/0022-006X.71.5.862

[5] Timothy W. Bickmore, Daniel Schulman, and Candace Sidner. 2013. Automated interventions for multiple health behaviors using conversational agents. *Patient Education and Counseling* 92, 2 (Aug. 2013), 142–148. doi:10.1016/j.pec.2013.05.011

[6] Timothy W. Bickmore, Daniel Schulman, and Candace L. Sidner. 2011. A reusable framework for health counseling dialogue systems based on a behavioral medicine ontology. *Journal of Biomedical Informatics* 44, 2 (April 2011), 183–197. doi:10.1016/j.jbi.2010.12.006

[7] Lois Biener and David B. Abrams. 1991. The Contemplation Ladder: Validation of a measure of readiness to consider smoking cessation. *Health Psychology* 10, 5 (1991), 360–365. doi:10.1037/0278-6133.10.5.360

[8] Edward S. Bordin. 1979. The generalizability of the psychoanalytic concept of the working alliance. *Psychotherapy: Theory, Research & Practice* 16, 3 (1979), 252–260. doi:10.1037/h0085885

[9] Virginia Braun, , and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (Jan. 2006), 77–101. doi:10.1191/1478088706qp063oa

[10] Ryan Broderick. 2023. People are using AI for therapy, whether the technology is ready for it or not. https://www.fastcompany.com/90836906/ai-therapy-koko-chatgpt [Accessed: 2025-11-12].

[11] Andrew Brown, Ash Tanuj Kumar, Osnat Melamed, Imtihan Ahmed, Yu Hao Wang, Arnaud Deza, Marc Morcos, Leon Zhu, Marta Maslej, Nadia Minian, Vidya Sujaya, Jodi Wolff, Olivia Doggett, Mathew Iantorno, Matt Ratto, Peter Selby, and Jonathan Rose. 2023. A Motivational Interviewing Chatbot With Generative Reflections for Increasing Readiness to Quit Smoking: Iterative Development Study. *JMIR Mental Health* 10 (Oct. 2023), e49132. doi:10.2196/49132

[12] Yu Ying Chiu, Ashish Sharma, Inna Wanyin Lin, and Tim Althoff. 2024. A Computational Framework for Behavioral Assessment of LLM Therapists. http://arxiv.org/abs/2401.00820 arXiv:2401.00820 [cs].

[13] Andrea Cuadra, Maria Wang, Lynn Andrea Stein, Malte F. Jung, Nicola Dell, Deborah Estrin, and James A. Landay. 2024. The Illusion of Empathy? Notes on Displays of Emotion in Human-Computer Interaction. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–18. doi:10.1145/3613904.3642336

[14] Simon D'Alfonso, Reeva Lederman, Sandra Bucci, and Katherine Berry. 2020. The Digital Therapeutic Alliance and Human-Computer Interaction. *JMIR Mental Health* 7, 12 (Dec. 2020), e21895. doi:10.2196/21895

[15] Yang Deng, Wenqiang Lei, Wai Lam, and Tat-Seng Chua. 2023. A Survey on Proactive Dialogue Systems: Problems, Methods, and Prospects. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*. International Joint Conferences on Artificial Intelligence Organization, Macau, SAR China, 6583–6591. doi:10.24963/ijcai.2023/738

[16] Bosheng Ding, Chengwei Qin, Linlin Liu, Yew Ken Chia, Boyang Li, Shafiq Joty, and Lidong Bing. 2023. Is GPT-3 a Good Data Annotator?. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Toronto, Canada, 11173–11195. https://aclanthology.org/2023.acl-long.626

[17] Geoffrey B. Duggan, Edmund Keogh, Gail A. Mountain, Paul McCullagh, Jason Leake, and Christopher Eccleston. 2015. Qualitative evaluation of the SMART2 self-management system for people in chronic pain. *Disability and Rehabilitation: Assistive Technology* 10, 1 (Jan. 2015), 53–60. doi:10.3109/17483107.2013.845696

[18] The Economist. 2025. Millions are turning to AI for therapy. https://www.economist.com/science-and-technology/2025/11/11/the-promise-and-the-perils-of-using-ai-for-therapy [Accessed: 2025-11-18].

[19] Robert Elliott, Arthur C. Bohart, Jeanne C. Watson, and David Murphy. 2018. Therapist empathy and client outcome: An updated meta-analysis. *Psychotherapy* 55, 4 (Dec. 2018), 399–410. doi:10.1037/pst0000175

[20] David P. Forman, Theresa B. Moyers, and Jon M. Houck. 2022. What can clients tell us about whether to use motivational interviewing? An analysis of early-session ambivalent language. *Journal of Substance Abuse Treatment* 132 (Jan. 2022), 108642. doi:10.1016/j.jsat.2021.108642

[21] Joana Galvão Gomes Da Silva, David J Kavanagh, Tony Belpaeme, Lloyd Taylor, Konna Beeson, and Jackie Andrade. 2018. Experiences of a Motivational Interview Delivered by a Robot: Qualitative Study. *Journal of Medical Internet Research* 20, 5 (May 2018), e116. doi:10.2196/jmir.7737

[22] Jacques Gaume, Molly Magill, Nadine R. Mastroleo, Richard Longabaugh, Nicolas Bertholet, Gerhard Gmel, and Jean-Bernard Daeppen. 2016. Change Talk During Brief Motivational Intervention With Young Adult Males: Strength Matters. *Journal of Substance Abuse Treatment* 65 (June 2016), 58–65. doi:10.1016/j.jsat.2016.01.005

[23] Shixian Geng, Remi Inayoshi, Chi-Lan Yang, Zefan Sramek, Yuya Umeda, Chiaki Kasahara, Arissa J. Sato, Simo Hosio, and Koji Yatani. 2025. Beyond the Dialogue: Multi-chatbot Group Motivational Interviewing for Premenstrual Syndrome (PMS) Management. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–18. doi:10.1145/3706598.3713918

[24] Sina Gessnitzer and Simone Kauffeld. 2015. The Working Alliance in Coaching: Why Behavior Is the Key to Success. *The Journal of Applied Behavioral Science* 51, 2 (June 2015), 177–197. doi:10.1177/0021886315576407

[25] Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. 2023. ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences* 120, 30 (July 2023), e2305016120. doi:10.1073/pnas.2305016120

[26] Anthony M. Grant. 2014. Autonomy support, relationship satisfaction and goal focus in the coach–coachee relationship: which best predicts coaching success? *Coaching: An International Journal of Theory, Research and Practice* 7, 1 (Jan. 2014), 18–38. doi:10.1080/17521882.2013.850106

[27] Carolin Graßmann, Franziska Schölmerich, and Carsten C Schermuly. 2020. The relationship between working alliance and client outcomes in coaching: A meta-analysis. *Human Relations* 73, 1 (Jan. 2020), 35–58. doi:10.1177/0018726718819725

[28] Itika Gupta, Barbara Di Eugenio, Brian D. Ziebart, Bing Liu, Ben S. Gerber, and Lisa K. Sharp. 2021. Summarizing Behavioral Change Goals from SMS Exchanges to Support Health Coaches. In *Proceedings of the 22nd Annual Meeting of the Special Interest Group on Discourse and Dialogue*. Association for Computational Linguistics, Singapore and Online, 276–289. https://aclanthology.org/2021.sigdial-1.31

[29] Juan Martín Gómez Penedo, Anna Margarete Babl, Martin Grosse Holtforth, Fritz Hohagen, Tobias Krieger, Wolfgang Lutz, Björn Meyer, Steffen Moritz, Jan Philipp Klein, and Thomas Berger. 2020. The Association of Therapeutic Alliance With Long-Term Outcome in a Guided Internet Intervention for Depression: Secondary Analysis From a Randomized Control Trial. *Journal of Medical Internet Research* 22, 3 (March 2020), e15824. doi:10.2196/15824

[30] Robert L. Hatcher and J. Arthur Gillaspy. 2006. Development and validation of a revised short version of the working alliance inventory. *Psychotherapy Research* 16, 1 (Jan. 2006), 12–25. doi:10.1080/10503300500352500

[31] Linwei He, Erkan Basar, Emiel Krahmer, Reinout Wiers, and Marjolijn Antheunis. 2024. Effectiveness and User Experience of a Smoking Cessation Chatbot: Mixed Methods Study Comparing Motivational Interviewing and Confrontational Counseling. *Journal of Medical Internet Research* 26 (Aug. 2024), e53134. doi:10.2196/53134

[32] Xingwei He, Zhenghao Lin, Yeyun Gong, A.-Long Jin, Hang Zhang, Chen Lin, Jian Jiao, Siu Ming Yiu, Nan Duan, and Weizhu Chen. 2023. AnnoLLM: Making Large Language Models to Be Better Crowdsourced Annotators. http://arxiv.org/abs/2303.16854 arXiv:2303.16854 [cs].

[33] Leighanne Hsu, Rommy Marquez Hernandez, Kathleen McCoy, Keith Decker, Ajith Vemuri, Greg Dominick, and Megan Heintzelman. 2022. Towards Development of an Automated Health Coach. In *Proceedings of the First Workshop on Natural Language Generation in Healthcare*. Association for Computational Linguistics, Waterville, Maine, USA and virtual meeting, 27–39. https://aclanthology.org/2022.nlg4health-1.4

[34] Fan Huang, Haewoon Kwak, and Jisun An. 2023. Is ChatGPT better than Human Annotators? Potential and Limitations of ChatGPT in Explaining Implicit Hate Speech. In *Companion Proceedings of the ACM Web Conference 2023*. ACM, Austin TX USA, 294–297. doi:10.1145/3543873.3587368

[35] Matthew Jörke, Shardul Sapkota, Lyndsea Warkenthien, Niklas Vainio, Paul Schmiedmayer, Emma Brunskill, and James A. Landay. 2025. GPTCoach: Towards LLM-Based Physical Activity Coaching. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–46. doi:10.1145/3706598.3713819

[36] Toshikazu Kanaoka and Bilge Mutlu. 2015. Designing a Motivational Agent for Behavior Change in Physical Activity. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems*. ACM, Seoul Republic of Korea, 1445–1450. doi:10.1145/2702613.2732924

[37] Dongjin Kang, Sunghwan Kim, Taeyoon Kwon, Seungjun Moon, Hyunsouk Cho, Youngjae Yu, Dongha Lee, and Jinyoung Yeo. 2024. Can Large Language Models be Good Emotional Supporter? Mitigating Preference Bias on Emotional Support Conversation. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 15232–15261. doi:10.18653/v1/2024.acl-long.813

[38] Siyang Liu, Chujie Zheng, Orianna Demasi, Sahand Sabour, Yu Li, Zhou Yu, Yong Jiang, and Minlie Huang. 2021. Towards Emotional Support Dialog Systems. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli (Eds.). Association for Computational Linguistics, Online, 3469–3483. doi:10.18653/v1/2021.acl-long.269

[39] Michael B. Madson, Richard S. Mohn, Julie A. Schumacher, and Alicia S. Landry. 2015. Measuring Client Experiences of Motivational Interviewing During a Lifestyle Intervention. *Measurement and Evaluation in Counseling and Development* 48, 2 (April 2015), 140–151. doi:10.1177/0748175614544687

[40] Michael B. Madson, Richard S. Mohn, Allan Zuckoff, Julie A. Schumacher, Jane Kogan, Shari Hutchison, Emily Magee, and Bradley Stein. 2013. Measuring client perceptions of motivational interviewing: factor analysis of the Client Evaluation of Motivational Interviewing scale. *Journal of Substance Abuse Treatment* 44, 3 (March 2013), 330–335. doi:10.1016/j.jsat.2012.08.015

[41] Michael B. Madson, Margo C. Villarosa, Julie A. Schumacher, and Richard S. Mohn. 2016. Evaluating the Validity of the Client Evaluation of Motivational Interviewing Scale in a Brief Motivational Intervention for College Student Drinkers. *Journal of Substance Abuse Treatment* 65 (June 2016), 51–57. doi:10.1016/j.jsat.2016.02.001

[42] Carol Ann Maher, Courtney Rose Davis, Rachel Grace Curtis, Camille Elizabeth Short, and Karen Joy Murphy. 2020. A Physical Activity and Diet Program Delivered by Artificially Intelligent Virtual Health Coach: Proof-of-Concept Study. *JMIR mHealth and uHealth* 8, 7 (July 2020), e17558. doi:10.2196/17558

[43] Tobias Mayer, Neha Warikoo, Amir Eliassaf, Dana Atzil-Slonim, and Iryna Gurevych. 2024. Predicting Client Emotions and Therapist Interventions in Psychotherapy Dialogues. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, Yvette Graham and Matthew Purver (Eds.). Association for Computational Linguistics, St. Julian's, Malta, 1463–1477. https://aclanthology.org/2024.eacl-long.88

[44] José Mercado, Ismael Edrein Espinosa-Curiel, and Juan Martínez-Miranda. 2023. Embodied Conversational Agents Providing Motivational Interviewing to Improve Health-Related Behaviors: Scoping Review. *Journal of Medical Internet Research* 25 (Dec. 2023), e52097. doi:10.2196/52097

[45] Selina Meyer and David Elsweiler. 2024. "You tell me": A Dataset of GPT-4-Based Behaviour Change Support Conversations. In *Proceedings of the 2024 ACM SIGIR Conference on Human Information Interaction and Retrieval*. ACM, Sheffield United Kingdom, 411–416. doi:10.1145/3627508.3638330

[46] Selina Meyer and David Elsweiler. 2025. LLM-based conversational agents for behaviour change support: A randomised controlled trial examining efficacy, safety, and the role of user behaviour. *International Journal of Human-Computer Studies* 200 (May 2025), 103514. doi:10.1016/j.ijhcs.2025.103514

[47] William R. Miller and Stephen Rollnick. 2023. *Motivational interviewing: helping people change and grow* (fourth edition ed.). The Guilford Press, New York.

[48] T. B. Moyers, J. K. Manuel, and D. Ernst. 2014. Motivational Interviewing Treatment Integrity Coding Manual 4.1.

[49] Tim Norfolk, Kamal Birdi, and Deirdre Walsh. 2007. The role of empathy in establishing rapport in the consultation: a new model. *Medical Education* 41, 7 (July 2007), 690–697. doi:10.1111/j.1365-2923.2007.02789.x

[50] Stefan Olafsson, Paola Pedrelli, Byron C. Wallace, and Timothy Bickmore. 2023. Accomodating User Expressivity while Maintaining Safety for a Virtual Alcohol Misuse Counselor. In *Proceedings of the 23rd ACM International Conference on Intelligent Virtual Agents*. ACM, Würzburg Germany, 1–9. doi:10.1145/3570945.3607361

[51] Stefan Olafsson, Byron Wallace, and Timothy Bickmore. 2020. Towards a Computational Framework for Automating Substance Use Counseling with Virtual Agents. In *Proceedings of the 19th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS '20)*. International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 966–974. https://dl.acm.org/doi/10.5555/3398761.3398874

[52] Jeanette M. Olsen. 2014. Health Coaching: A Concept Analysis: Health Coaching Concept Analysis. *Nursing Forum* 49, 1 (Jan. 2014), 18–29. doi:10.1111/nuf.12042

[53] Jeanette M. Olsen and Bonnie J. Nesbitt. 2010. Health Coaching to Improve Healthy Lifestyle Behaviors: An Integrative Review. *American Journal of Health Promotion* 25, 1 (Sept. 2010), e1–e12. doi:10.4278/ajhp.090313-LIT-101

[54] SoHyun Park, Jeewon Choi, Sungwoo Lee, Changhoon Oh, Changdai Kim, Soohyun La, Joonhwan Lee, and Bongwon Suh. 2019. Designing a Chatbot for a Brief Motivational Interview on Stress Management: Qualitative Case Study. *Journal of Medical Internet Research* 21, 4 (April 2019), e12231. doi:10.2196/12231

[55] Havisha Pedamallu, Matthew J Ehrhardt, Julia Maki, April Idalski Carcone, Melissa M Hudson, and Erika A Waters. 2022. Technology-Delivered Adaptations of Motivational Interviewing for the Prevention and Management of Chronic Diseases: Scoping Review. *Journal of Medical Internet Research* 24, 8 (Aug. 2022), e35283. doi:10.2196/35283

[56] Corina Pelau, Dan-Cristian Dabija, and Irina Ene. 2021. What makes an AI device human-like? The role of interaction quality, empathy and perceived psychological anthropomorphic characteristics in the acceptance of artificial intelligence in the service industry. *Computers in Human Behavior* 122 (Sept. 2021), 106855. doi:10.1016/j.chb.2021.106855

[57] Julie Prescott and Terry Hanley. 2023. Therapists' attitudes towards the use of AI in therapeutic practice: considering the therapeutic alliance. *Mental Health and Social Inclusion* 27, 2 (Jan. 2023), 177–185. doi:10.1108/MHSI-02-2023-0020

[58] James O. Prochaska and Wayne F. Velicer. 1997. The Transtheoretical Model of Health Behavior Change. *American Journal of Health Promotion* 12, 1 (Sept. 1997), 38–48. doi:10.4278/0890-1171-12.1.38

[59] Eugénia Ribeiro, António P. Ribeiro, Miguel M. Gonçalves, Adam O. Horvath, and William B. Stiles. 2013. How collaboration in therapy becomes therapeutic: The therapeutic collaboration coding system. *Psychology and Psychotherapy: Theory, Research and Practice* 86, 3 (Sept. 2013), 294–314. doi:10.1111/j.2044-8341.2012.02066.x

[60] Nicole L. Robinson, Jennifer Connolly, Leanne Hides, and David J. Kavanagh. 2020. Social robots as treatment agents: Pilot randomized controlled trial to deliver a behavior change intervention. *Internet Interventions* 21 (Sept. 2020),

100320. doi:10.1016/j.invent.2020.100320

[61] Heleen Rutjes, Martijn C. Willemsen, and Wijnand A. IJsselsteijn. 2019. Beyond Behavior: The Coach's Perspective on Technology in Health Coaching. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, Glasgow Scotland Uk, 1–14. doi:10.1145/3290605.3300900

[62] Lennart Seitz. 2024. Artificial empathy in healthcare chatbots: Does it feel authentic? *Computers in Human Behavior: Artificial Humans* 2, 1 (Jan. 2024), 100067. doi:10.1016/j.chbah.2024.100067

[63] Ashish Sharma, Adam Miner, David Atkins, and Tim Althoff. 2020. A Computational Approach to Understanding Empathy Expressed in Text-Based Mental Health Support. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, Online, 5263–5276. doi:10.18653/v1/2020.emnlp-main.425

[64] Vishal Anand Shetty, Shauna Durbin, Meghan S Weyrich, Airín Denise Martínez, Jing Qian, and David L Chin. 2024. A scoping review of empathy recognition in text using natural language processing. *Journal of the American Medical Informatics Association* 31, 3 (Feb. 2024), 762–775. doi:10.1093/jamia/ocad229

[65] Rebecca M. Shingleton and Tibor P. Palfai. 2016. Technology-delivered adaptations of motivational interviewing for health-related behaviors: A systematic review of the current research. *Patient Education and Counseling* 99, 1 (Jan. 2016), 17–35. doi:10.1016/j.pec.2015.08.005

[66] Ben Singh, Timothy Olds, Jacinta Brinsley, Dot Dumuid, Rosa Virgara, Lisa Matricciani, Amanda Watson, Kimberley Szeto, Emily Eglitis, Aaron Miatke, Catherine E. M. Simpson, Corneel Vandelanotte, and Carol Maher. 2023. Systematic review and meta-analysis of the effectiveness of chatbots on lifestyle behaviours. *npj Digital Medicine* 6, 1 (June 2023), 118. doi:10.1038/s41746-023-00856-1

[67] James D. Slavet, L.A.R. Stein, Suzanne M. Colby, Nancy P. Barnett, Peter M. Monti, Charles Golembeske, and Rebecca Lebeau-Craven. 2006. The Marijuana Ladder: Measuring motivation to change marijuana use in incarcerated adolescents. *Drug and Alcohol Dependence* 83, 1 (June 2006), 42–48. doi:10.1016/j.drugalcdep.2005.10.007

[68] Dave Smits, Koen Luyckx, Dirk Smits, Nele Stinckens, and Laurence Claes. 2015. Structural characteristics and external correlates of the Working Alliance Inventory-Short Form. *Psychological Assessment* 27, 2 (June 2015), 545–551. doi:10.1037/pas0000066

[69] Vidya Srinivas, Xuhai Xu, Xin Liu, Kumar Ayush, Isaac Galatzer-Levy, Shwetak Patel, Daniel McDuff, and Tim Althoff. 2025. Substance over Style: Evaluating Proactive Conversational Coaching Agents. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (Eds.). Association for Computational Linguistics, Vienna, Austria, 20848–20880. doi:10.18653/v1/2025.acl-long.1017

[70] Elizabeth C. Stade, Shannon Wiltsey Stirman, Lyle H. Ungar, Cody L. Boland, H. Andrew Schwartz, David B. Yaden, João Sedoc, Robert J. DeRubeis, Robb Willer, and Johannes C. Eichstaedt. 2024. Large language models could change the future of behavioral healthcare: a proposal for responsible development and evaluation. *npj Mental Health Research* 3, 1 (April 2024), 12. doi:10.1038/s44184-024-00056-z

[71] Ian Steenstra, Farnaz Nouraei, Mehdi Arjmand, and Timothy Bickmore. 2024. Virtual Agents for Alcohol Use Counseling: Exploring LLM-Powered Motivational Interviewing. In *Proceedings of the ACM International Conference on Intelligent Virtual Agents*. ACM, GLASGOW United Kingdom, 1–10. doi:10.1145/3652988.3673932

[72] Xin Sun, Xiao Tang, Abdallah El Ali, Zhuying Li, Pengjie Ren, Jan de Wit, Jiahuan Pei, and Jos A.Bosch. 2025. Rethinking the Alignment of Psychotherapy Dialogue Generation with Motivational Interviewing Strategies. In *Proceedings of the 31st International Conference on Computational Linguistics*, Owen Rambow, Leo Wanner, Marianna Apidianaki, Hend Al-Khalifa, Barbara Di Eugenio, and Steven Schockaert (Eds.). Association for Computational Linguistics, Abu Dhabi, UAE, 1983–2002. https://aclanthology.org/2025.coling-main.136/

[73] Sara Syed, Zainab Iftikhar, Amy Wei Xiao, and Jeff Huang. 2024. Machine and Human Understanding of Empathy in Online Peer Support: A Cognitive Behavioral Approach. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–13. doi:10.1145/3613904.3642034

[74] Quyen G To, Chelsea Green, and Corneel Vandelanotte. 2021. Feasibility, Usability, and Effectiveness of a Machine Learning–Based Physical Activity Chatbot: Quasi-Experimental Study. *JMIR mHealth and uHealth* 9, 11 (Nov. 2021), e28577. doi:10.2196/28577

[75] Bruce E. Wampold and Zac E. Imel. 2015. *The great psychotherapy debate: the evidence for what makes psychotherapy work* (second edition ed.). Routledge, Taylor & Francis, New York.

[76] Henny A. Westra and Nikoo Norouzian. 2018. Using Motivational Interviewing to Manage Process Markers of Ambivalence and Resistance in Cognitive Behavioral Therapy. *Cognitive Therapy and Research* 42, 2 (April 2018), 193–203. doi:10.1007/s10608-017-9857-6

[77] Zixiu Wu, Simone Balloccu, Vivek Kumar, Rim Helaoui, Diego Reforgiato Recupero, and Daniele Riboni. 2023. Creation, Analysis and Evaluation of AnnoMI, a Dataset of Expert-Annotated Counselling Dialogues. *Future Internet* 15, 3 (March 2023), 110. doi:10.3390/fi15030110

[78] Bo Xiao, Dogan Can, Panayiotis G. Georgiou, David Atkins, and Shrikanth S. Narayanan. 2012. Analyzing the Language of Therapist Empathy in Motivational Interview based Psychotherapy. *Signal and Information Processing Association Annual Summit and Conference (APSIPA), ... Asia-Pacific. Asia-Pacific Signal and Information Processing Association Annual Summit and Conference* 2012 (Dec. 2012), 6411762.

[79] Zhouhang Xie, Bodhisattwa Prasad Majumder, Mengjie Zhao, Yoshinori Maeda, Keiichi Yamada, Hiromi Wakaki, and Julian McAuley. 2024. Few-shot Dialogue Strategy Learning for Motivational Interviewing via Inductive Reasoning. In *Findings of the Association for Computational Linguistics ACL 2024*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand and virtual meeting, 13207–13219. doi:10.18653/v1/2024.findings-acl.782

A Data

We utilise the AnnoMI [77] dataset as inputs to customise the prompts for the motivation detector (Section 3.1), the strategy selector (Section 3.3), and the response generator (Section 3.4). It consists of 133 MI conversations available under Public Domain License in 10 different topics in English which are transcribed from YouTube demonstration videos and annotated by experts from the MI network[13]. Post-annotation surveys revealed that most annotators considered these videos representative of authentic MI sessions, despite their scripted nature for educational purposes.

An utterance may comprise multiple sentences, with labels assigned at the turn level rather than the sentence level. Client utterances serve as the data for detecting client motivational language (Section 3.1) while counsellor utterances are utilised as the database for predicting appropriate coaching strategies for agent responses (Sections 3.3 and 3.4). The following two sub-sections provide detailed descriptions of both client and counsellor utterances.

A.1 Client Utterance

Detecting client motivational language requires two annotation labels: `attitude` and `strength`. While each client utterance in AnnoMI has already been assigned an `attitude` label ("change", "sustain", or "neutral"), they lack `strength` labels. Therefore, we undertook the manual annotation process to add the missing `strength` labels to all client utterances with assistance from GPT models.

Recent research on LLMs in data annotation [16, 25, 32, 34] suggests they can assist annotators by providing detailed explanations, potentially replacing crowdsourced workers. MI Skill Code (MISC)[14] is a behavioural coding system for MI sessions. Using the MISC guidelines and the explanations generated by GPT-3.5, the first author annotated client turns with strength values (i.e., "high", "medium", or "low"). We employed the strength rating approach similar to that of Gaume et al. [22]. Each client utterance is first assigned a strength value of "medium". If the utterance contains a booster word (e.g., "definitely", "for sure"), its strength value changes to "high". On the contrary, if it has one or more hedge words (e.g., "maybe", "I think"), it receives "low" value. The "neutral" type is originally not assigned a strength value but, we decided to annotate it similarly to the other two types for the sake of completeness. Since textual information alone is insufficient, we watched corresponding videos from `AnnoMI` dataset and then use speech signals and body languages to assist with the annotation process.

A.2 Counsellor Utterance

The counsellor utterances in AnnoMI are labelled using an annotation scheme that slightly differs from standard MI labels. To enable evaluation of the model outputs using established MI metrics, the first author relabelled these utterances according to the Motivational Interviewing Treatment Integrity (MITI 4.2.1) manual [48]. The relabelling was performed using the Claude service[15]. Table 2 presents the MI-adherent strategies employed in our study, either to build rapport or push towards goal change, together with their descriptions.

[13]https://motivationalinterviewing.org/
[14]https://casaa.unm.edu/tools/misc.html
[15]https://claude.ai/

B Qualitative Coding on Behavioural Dimensions

B.1 Annotation Guidelines

Since this deductive thematic analysis employed fixed themes, our main task was to identify sub-codes that best represented our data. We randomly selected 10% of conversations for coding. The first author documented potential sub-themes in Excel, with each column representing one sub-theme per conversation. Subsequently, we consulted papers analysing elements of digital therapeutic alliance [14] and identifying user behaviours in coaching contexts [46]. We then identified overlapping sub-themes and those occurring in the majority of coded conversations. Through this process, three behavioral dimensions emerged organically. The first author then developed an annotation guideline and coded all conversations accordingly using a binary classification system. The detailed annotation guidelines for each behavioural dimension are as follows:

- Plan Agreement ("Goal/Task" Theme): The coaching agents collaborate with users to develop specific, concrete change plans by the end of each session. Specific plans include information such as the agreed-upon activities, the timing/frequency, and the duration (e.g., walking 3 times a week). Conversely, sessions that fail to reach agreement on plans occur when users do not explicitly agree to proposed actions or when the plan lacks specificity.
- Proactivity ("Goal/Task" Theme): Proactive users engage in developing their exercise plans. Rather than seeking quick fixes or expecting ready-made solutions, they work collaboratively with the AI coaches. They may occasionally ask for suggestions, but contribute their own thoughts and ideas throughout the process. Non-proactive users, in contrast, rely heavily on the AI coaches for direction, frequently asking for suggestions and recommendations while contributing minimal input themselves. They may offer occasional input but generally defer to the coaches' ideas.
- Cooperation ("Bond" Theme): Cooperative users consistently reply in full sentences and makes an effort to respond to the AI coaches, answering questions thoughtfully and contributing meaningfully to the conversations. In contrast, uncooperative users tend to provide brief, minimal responses with little effort to engage meaningfully. They may also offer random or off-topic responses, redirect the conversation to irrelevant subjects, or abruptly end sessions immaturely.

To validate the annotations, we employed GPT-4o as an independent theme annotator. We provided the model with the annotation guideline and the conversations, instructing it to apply the binary coding scheme to each conversation according to the established criteria. We then compared our annotations with those generated by GPT-4o. In cases of coding discrepancies, we carefully reviewed both the model's reasoning and the annotation guideline before determining whether to retain or revise the original coding.

B.2 Chat Examples

B.2.1 User 107, MI-aware condition. Dimensions: plan agreement: 0, proactivity: 1, cooperation: 1.

User: Bunny

Jordan: Hi Bunny, I'm Jordan, your physical activity coach. My role is to support you in becoming more active. Could you share a bit about yourself and what brings you here today?

User: Hi, I'm a middle aged male, currently doing next to do physical exercise, but would like to feel a bit fitter.

Jordan: What do you hope to achieve in our session today, Bunny?

User: I'm not sure, perhaps some additional motivation to find time to exercise.

Jordan: It sounds like finding time for exercise is a challenge for you. Could you share more about what motivates you to want to feel fitter?

User: Just the usual passage of time and the way it impacts your body. While I remain fairly able at the moment, I'm starting to notice the odd discomfort and the like and I'd like to forestall that as much as possible.

Jordan: It seems you're noticing changes and want to maintain your abilities. What types of activities do you find enjoyable or fulfilling?

User: I quite enjoy walking.

Jordan: Walking is something you enjoy. How often do you currently go for walks, and what do you like most about them?

User: I used to go moderately frequently, walking once a day for groceries, and hour-long walks once a week. I enjoy getting out in the fresh air, and the stimulation of being outside with all its activity.

Jordan: What about walking now makes you hesitant or self-conscious? Being outside and active seems to bring you joy and energy, yet something might be holding you back from fully engaging in it.

User: The current issue is that I'm having to temporarily host my father for a couple of months, and the only time I really get to myself is when he pops out briefly, so I'm loathe to lose that time by heading out on a walk.

Jordan: On a scale from 1 to 10, how important is it for you to find time for walking despite your current situation?

User: Perhaps a 6

Jordan: What specific steps do you think you could take to fit in some walking during your day? How often would you like to aim for, and how might you measure your progress?

User: While my dad's here, I'd like to make it more of a regular routine, maybe once every couple of days.

Jordan: Creating a routine can be beneficial. Walking can improve cardiovascular health and boost mood. Consider setting a goal, like a 20-minute walk every other day, and track your progress.

User: What would be the purpose of tracking it?

Jordan: Tracking your progress can help you see improvements over time and keep you motivated. How do you feel about using a simple log or app to record your walks? Remember, it's your choice how to proceed.

User: I'm not really looking to improve my walking ability though, it's not like other fitness activities such as lifting weights, it's just an enjoyable activity by itself.

Jordan: That's great! Enjoyment is a crucial part of staying active. You could still track it for the joy of seeing your consistency. If you're open to it, gradually increasing intensity, like adding short jogs, might keep it engaging. Your comfort and enjoyment are the priority, so feel free to adjust as you see fit.

User: I don't enjoy jogging, it's quite jarring on the body.

Jordan: Walking is excellent for health and can be done anywhere, not just at the gym. Even a few minutes daily can be beneficial. How might you incorporate more enjoyable walking into your routine?

User: I suppose I could accompany my dad on his walks, although I'd prefer to go solo.

Jordan: Walking solo can be a great way to unwind. Perhaps consider listening to music or podcasts. It can make the time fly and might even naturally increase your pace. How does that sound to you?

User: Oh it certainly increases the pace. I can tell when I've listened to a political podcast as I zip around at twice the pace, although it kind of kills the relaxation that comes with walking.

Jordan: It sounds like you enjoy the faster pace but also value relaxation. Maybe you could alternate between upbeat podcasts and calming music to balance the two. How do you feel about trying this approach?

User: We'll see, I might not even be doing the walk solo, in which case audio listening is no longer an option

Jordan: If you find yourself walking with someone, perhaps consider setting a conversational pace that feels comfortable but still energizes you. It's all about finding what works best for you. What do you think?

User: Perhaps that could work, we'll see. Bye

Jordan: Thank you for your openness during our session. I wish you success in finding the right balance on your walks. Goodbye and take care.

B.2.2 User 159, base-MI condition. Dimensions: plan agreement: 0, proactivity: 0, cooperation: 1

User: JEB

Jordan: How are you feeling about your current physical activity routine, JEB?

User: I feel pretty good about it, but I could do better. I think the problem with me is that I don't have the time to be more physically active than what I currently do

Jordan: I understand, JEB. Finding time can be challenging. What small changes do you think could help you incorporate a bit more activity into your routine?

User: Well I don't know! I'd have to sacrifice other things I love doing. I already go to the gym 2 or 3 times a week. My session lasts just under 2 hours.

Jordan: It sounds like you're committed to your gym routine. Is there a way to integrate brief activities into your day without sacrificing what you love?

User: If I could work whilst going for a walk I'd do that, but it's inefficient, and I'm not sure what the benefit is of walking really? Does it help that much?

Jordan: Walking can be quite beneficial, improving cardiovascular health and boosting mood. Perhaps starting with short walks could help you feel the benefits without feeling inefficient. How does that sound?

User: I'm not sure you understood me. What I mean is that when I go for a walk how much benefit am I getting out of the walk vs time sacrificed. If I walk for an hour, a large portion of my day is gone after that hour. Sometimes I go for walks and do stuff on my phone to get work done at the same time, but the work is done to the same quality or focus

Jordan: I hear your concern about time efficiency, JEB. How about trying shorter, more frequent walks to see if they offer a balance between activity and productivity for you?

User: I think that is more disruptive, better to take just one big walk. Unless you are saying I do short amounts but every single day instead of maybe an hour walk over multiple days

Jordan: That could be an option, JEB. Short daily walks might integrate more seamlessly into your routine, offering consistent benefits. Would you be open to experimenting with this approach?

User: I'd be open to trying it, but I am not satisfied with that idea really, I'll give it a shot

Jordan: It's great that you're willing to try. Let's see how it feels and adjust as needed. Your feedback will be important to find what truly works for you.

User: Thanks"

Jordan: You're welcome, JEB. Remember, it's a process, and we'll figure out the best fit together. Let me know how it goes, and we can make adjustments based on your experience.

B.2.3 User 122, MI-aware condition. Dimensions: plan agreement: 1, proactivity: 0, cooperation: 0

User: Tim

Jordan: Hi Tim, I'm Jordan, your physical activity coach. My role is to support you in becoming more active. Could you tell me a bit about yourself and what brings you here today?

User: I want advice to get more exercise

Jordan: What would you like to achieve in our session today, Tim?

User: Easy ways to get moving

Jordan: Before we dive into strategies, let's explore your motivation. What led you to stop exercising before, and what other reasons might have contributed? You want easy ways to get moving.

User: Getting a desk job

Jordan: It's your choice to set the pace of your fitness journey. How do you feel your desk job affects your motivation or ability to stay active?

User: Yes

Jordan: It sounds like your desk job has an impact. What would you like to change about your current level of physical activity?

User: Get my heart rate up

Jordan: It's great that you're focusing on heart health. You're dedicated to improving your well-being. What motivates you to prioritize increasing your heart rate through physical activity?

User: To feel and look better

Jordan: On a scale from 1 to 10, how important is it for you to feel and look better through increased physical activity?

User: 8

Jordan: You're quite motivated, which is fantastic. Adults are encouraged to aim for 150 minutes of moderate activity weekly. How do you feel about incorporating this into your routine? Remember, your journey is yours to design based on your lifestyle and preferences.

User: I feel great about it

Jordan: It's wonderful that you're considering consulting your doctor to ensure your safety, showing a thoughtful approach to your health. What makes you feel it's an 8 and not, say, a 5 or 6?

User: Just really motivated

Jordan: That's great to hear! A gradual approach can be very effective. You might consider increasing your activity by 5 to 10 minutes every few weeks. This way, reaching 30 minutes most days feels manageable. How does that sound to you?

User: Great

Jordan: Now that we've readjusted your goal, how confident are you on a scale from 0 to 10 about achieving this gradual increase in activity?

User: 9

Jordan: That's fantastic confidence! You have the freedom to choose when and how to incorporate activities. Adding a brisk walk after meals can help. Feel free to adjust this to suit your schedule and preferences.

User: I will do that

Jordan: I agree. It sounds like a great plan to handle things through lifestyle changes. How about we outline a specific schedule for those walks? Remember, it's your choice to adapt as needed.

User: Sounds good

Jordan: What specific days and times do you think would work best for your walks? And how often would you like to fit them into your week to ensure consistency?

User: Weekdays

Jordan: Great choice! How about starting with 30-minute walks, three times a week? As you get comfortable, you might consider increasing the pace or duration. Feel free to adjust as suits you best.

User: Great

Jordan: You've expressed a strong motivation to feel and look better by increasing your physical activity, aiming for consistent weekday walks. Is this summary accurate? As we wrap up, can you confirm your change goal? Do you have any questions to discuss in our next session?

User: Not really

Jordan: I'm glad to hear you're set on your goal. Remember, I'm here for support if needed. Thank you for your cooperation. Wishing you success on this journey. Goodbye.

C Prompts

Prompt Construction as Input to the Response Generation Model
System Prompt
Your name is Jordan. You are a physical activity coach using motivational interviewing method. Your task is to explore the clients' perceptions about physical activity. Explore their current exercise habits, desires for change, and potential barriers to engaging in more physical activity. Do not ask them anything about their plan. If the client asks for advice, plans, recommendations, or strategies, tell them you would like to discuss about their motivation first.
User Prompt
Response using the following coaching strategies: do simple reflection, ask an open focusing question. *Do simple reflection:* Reflections capture and return to the client something that the client has said. Reflections add little or no meaning or emphasis to what the client has said. Simple reflections merely convey understanding or facilitate client/counselor exchanges. Simply repeating or rephrasing what the client has said qualifies as a Simple Reflection. They may identify very important or intense client emotions but do not go far beyond the original overt content of the client's statement. *Ask an open focusing question:* An open question is to gather information, understand, or elicit the client's story about their concerns, their reasons for change, etc. Generally, these begin with a question marker word: Who, What, Why, When, How, Where, etc. The question may also be phased in the imperative statement such as "Tell me..." Use the following clue for the response: *"Tell me more about that."* and try to make it fit into the context. If you have used the clue before, do not simply repeat it again but phrase it in a slightly different way. Response in a calm, collected, and empathetic manner like a professional coaching. Do not sound overexcited. Keep the answer concise and under 200 characters long. The output should be the therapist's response and nothing else.
Dialogue History
Here is the conversation so far: Client: I agree. I think I need to build on that foundation. I'm open to trying different types of exercises, but I want to make sure that whatever I choose fits into my lifestyle and feels enjoyable. What suggestions do you have for increasing my activity levels? Therapist: You're in control of your journey. Feel free to explore various activities and discover what resonates with you personally. Client: I appreciate that approach. I've been thinking about options like cycling or swimming, but I'm not sure how to start or incorporate those into my routine. What would you recommend for easing into those activities? Therapist: It sounds like you're considering cycling and swimming as new activities to include in your routine. Client: Yes, exactly. I'm curious about how to start with cycling and swimming without overwhelming myself. What are some manageable steps I can take to ease into these activities?

Table 10: An example of input prompt for phase "focusing". The generated response is *"It seems you're looking for ways to start cycling and swimming without feeling overwhelmed. Tell me more about your current routine and what feels manageable for you."*, reflecting both simple-reflection and open-question strategies.

Prompt For Annotating MI Strategies in Conversations.
Instruction: You are given a dialogue between the therapist and the client. The therapist is using motivation interviewing techniques. Predict what the therapist's actions are in the last utterance. Each sentence in the last utterance should have one action. Only choose the actions from the action list. ### Action list: 1. Give Information: to provide information to the client, explain ideas or concepts to the intervention, educate about a topic, or express a professional opinion on the client's habits or actions without persuading, advising, or warning. Positive Example: "The guidelines state that adults should aim for do at least 150 minutes of moderate intensity activity a week or 75 minutes of vigorous intensity activity a week." Negative Example: "From my professional experience, I think that going to cardiac rehab is the best choice for you." (Persuade without Permission) Negative Example: "You indicated during the assessment that you typically drink about 18 standard drinks per week. This far exceeds social drinking." (Confront) ... ### Dialogue to evaluate: dialogue The output should be a list of actions for the last therapist's utterance: [RESULT] names of the actions

Table 11: Prompt to annotate MI strategies in conversations.